CORE Econ Micro

Technology and incentives

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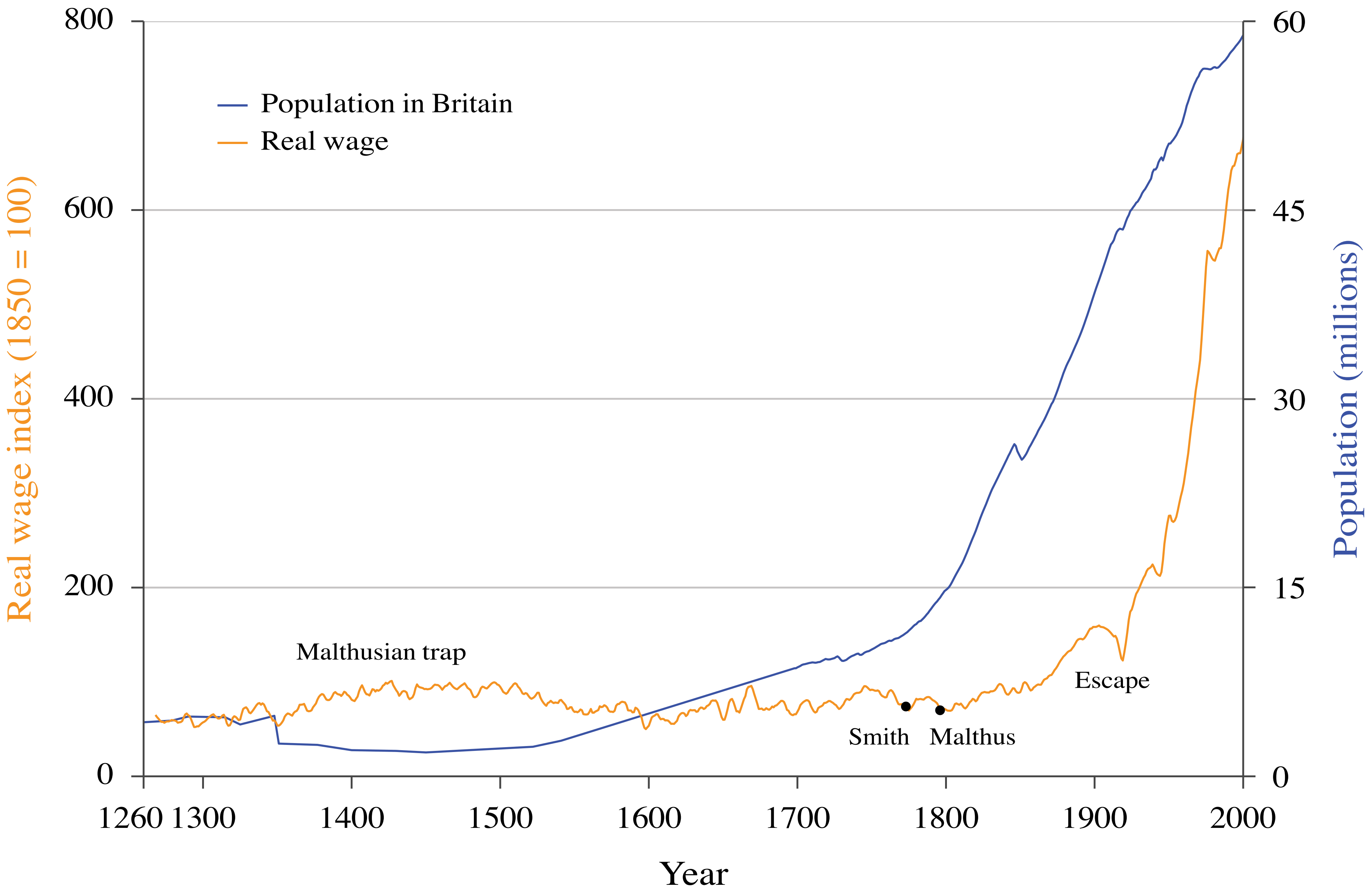
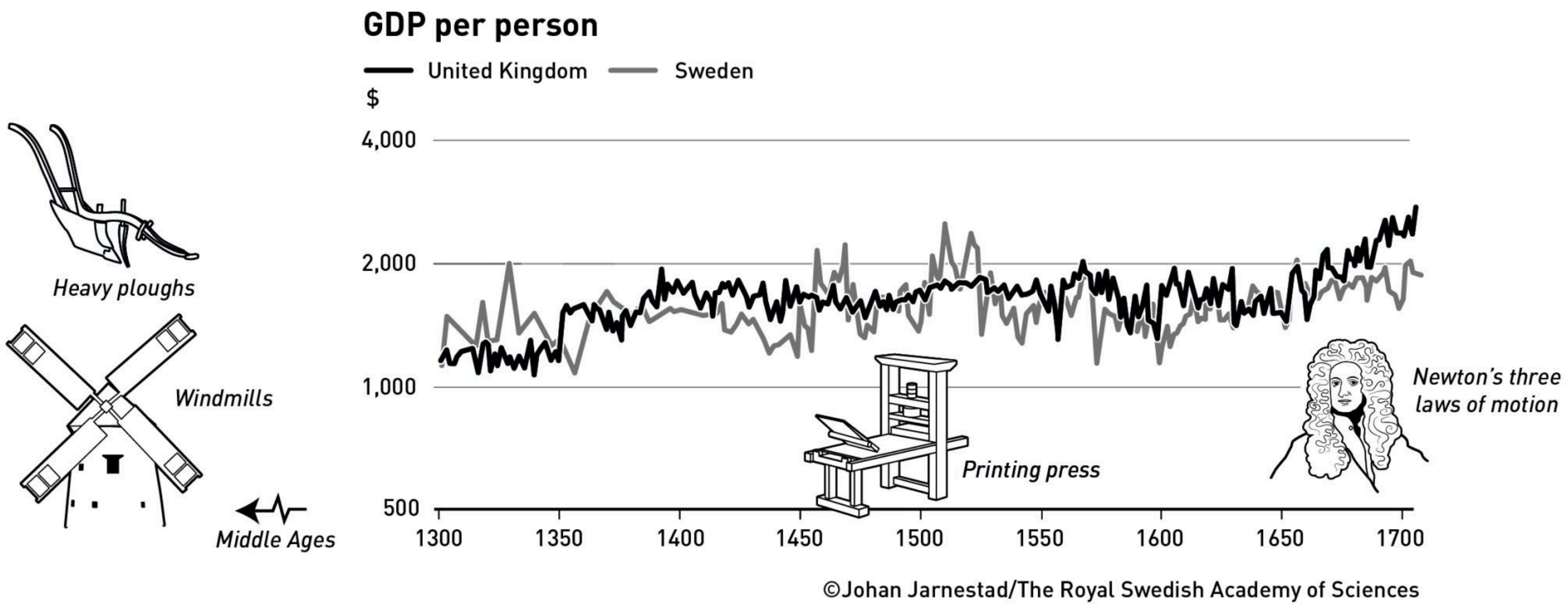

Population in Britain
Real wage
Malthusian trap
Smith
Malthus
Escape
Real wage index (1850 = 100)
Population (millions)
Year
800
600
400
200
0
60
45
30
15
0
1260
1300
1400
1500
1600
1700
1800
1900
2000



GDP per person
United Kingdom
Sweden
$
4,000
2,000
1,000
500
1300
1350
1400
1450
1500
1550
1600
1650
1700
Heavy ploughs
Windmills
Middle Ages
Printing press
Newton's three laws of motion
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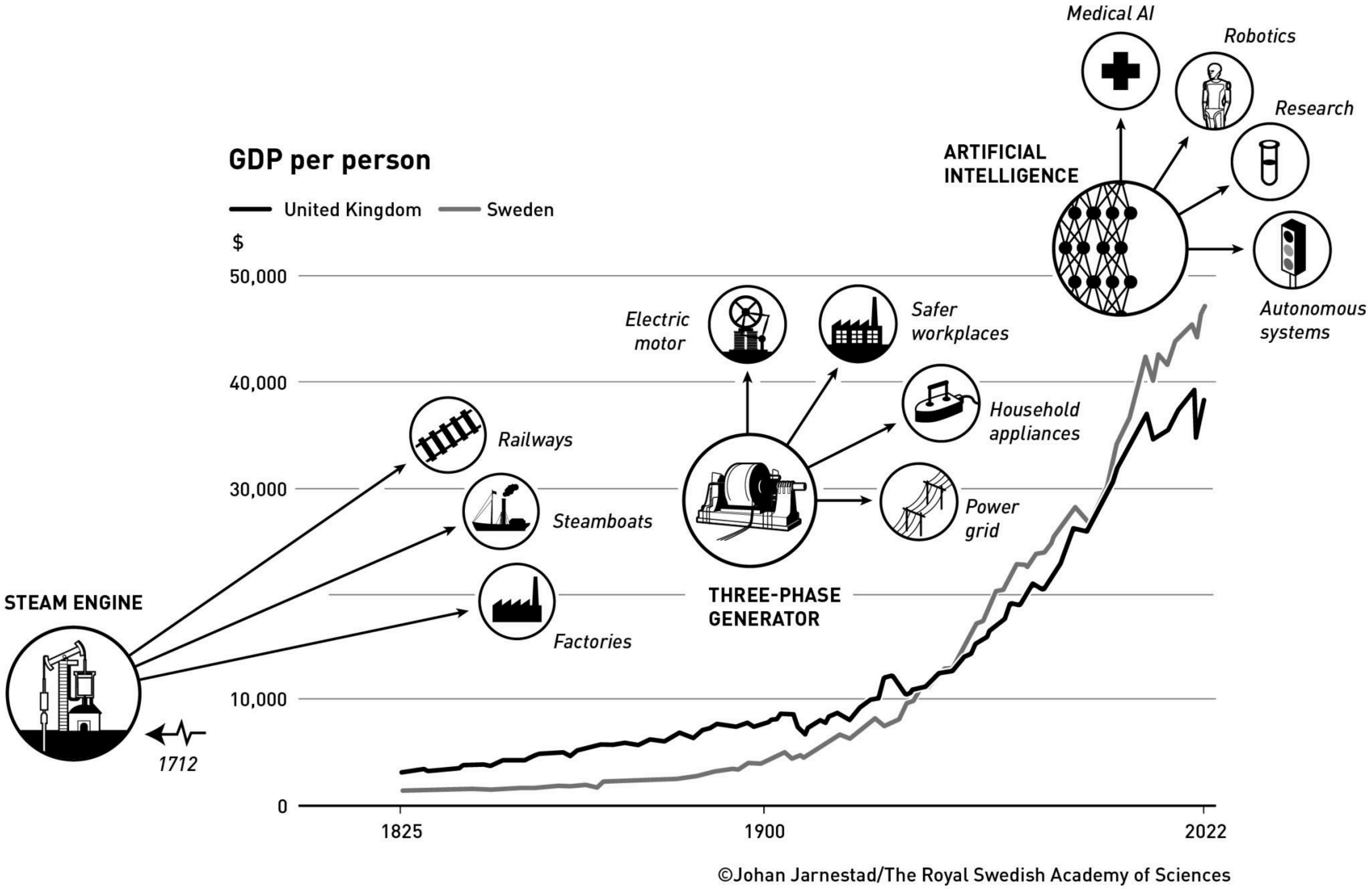

GDP per person
United Kingdom
Sweden
$
50,000
40,000
30,000
10,000
0
1825
1900
2022
STEAM ENGINE
1712
Railways
Steamboats
Factories
THREE-PHASE GENERATOR
Electric motor
Safer workplaces
Household appliances
Power grid
ARTIFICIAL INTELLIGENCE
Medical AI
Robotics
Research
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Economic decisions: Opportunity costs, economic rents, and incentives

Economic decisions: Opportunity costs, economic rents, and incentives

What happens in the economy is the result of decisions by individuals, firms (or their owners and managers), and governments.

We need to understand how people make decisions, and the factors they take into account when they choose what action to take.

Example #1: Concert

$$\begin{aligned}\text{net benefit of concert} &= \text{enjoyment from attending} - \text{cost of ticket}\\ &= \$55 - \$25\\ &= \$30\end{aligned}$$

Example #2: Babysitting

$$\begin{aligned}\text{net benefit of babysitting} &= \text{payment received} - \text{cost of effort involved}\\ &= \$40 - \$18\\ &= \$22\end{aligned}$$

Opportunity and Economic costs

Economic decisions often involve choosing between alternative and mutually exclusive courses of action.

By taking an action (concert) you lose the opportunity of taking the next best action (babysitting).

Opportunity cost: What you lose when you choose one action rather than the next best alternative. Also **reservation option**.

Actions have both direct costs and opportunity costs.

The economic cost of an action is the sum of these two components:

$$\text{economic cost of an action} = \text{direct costs incurred by taking the action} + \text{opportunity cost}$$

For the concert example:

$$\text{benefit} = \text{enjoyment of concert} = \$55$$

$$\text{economic cost} = \text{cost of ticket} + \text{opportunity cost} = \$25 + \$22 = \$47$$

What to do?

Opportunity and Economic costs

Another way to express the decision rule is: take the action if the net benefit is greater than the opportunity cost.

For the concert example:

$$\text{net benefit} = \text{enjoyment of concert} - \text{cost of ticket} = \$55 - \$25 = \$30$$

$$\text{opportunity cost} = \text{net benefit from babysitting} = \$22$$

Since the net benefit ($30) is greater than the opportunity cost ($22), you should go to the concert.

Economic rent: the difference between the net benefit (monetary or otherwise) that an individual receives from a chosen action, and the net benefit from the next best alternative

Incentives and Relative Prices

Economic rents exist throughout the economy and serve as incentives for people to take action.

For example, potential innovation rents may encourage firms to switch from one technology to another.

Economics focuses on alternatives and choices—people are free to select different actions, and their decisions depend on economic incentives such as potential rewards or penalties.

Typically, owners and managers choose the option that provides the highest profit, but other motivations—such as love, duty, ethics, or a desire for approval—can also influence decisions.

Relative prices: the price of one good or service compared to another (usually expressed as a ratio of the two prices).

A key determinant of economic incentives: they reflect the price of one option relative to another and guide decision-making for both firms and consumers.

If all prices rise by the same percentage (say 5%), the relative prices remain the same, and decisions are unlikely to change.

Comparative advantage, specialization, and markets

Comparative advantage, specialization, and markets

People do not typically produce the full range of goods and services that they use or consume in their daily lives.

We **specialize**:

- People specialize: some producing one good, others producing other goods; some working as welders, others as teachers or architects.
- Firms specialize: particular goods, or in just one part of the production process for a good.
- Countries specialize: Cambodia’s largest exports are gold and knitwear; Singapore specializes in integrated circuits.

Why?

- **Learning by doing**: We acquire skills as we produce things.
- **Difference in ability**: Because of skill, or the characteristics of the natural surroundings (such as soil quality, for example), some people are better at producing certain things than others.
- **Economies of scale**: Producing a large number of units of a good is often more cost-effective than producing a smaller number.

Comparative advantage: Gains from specialization

Specialization: each person does just one or a few of the large number of tasks necessary to complete some project or reach some goal.

Counter intuitive: all producers can benefit, even when this means that one producer specializes in a good that another could produce at lower cost.

	Production if 100% of time is spent on one good
Greta	1,250 apples or 50 tons of wheat
Carlos	1,000 apples or 20 tons of wheat

Absolute advantage: a situation where a person or country can produce more of a good than another person or country using the same amount of resources or inputs.

$$\# \, Apples_{Greta} = 1,250 > \# \, Apples_{Carlos} = 1000$$

$$\# \, Tons \; of \; wheat_{Greta} = 50 > \# \, Tons \; of \; wheat_{Carlos} = 20$$

$\Rightarrow$ Greta has absolute advantage producing both goods

Comparative advantage: Gains from specialization

Specialization: each person does just one or a few of the large number of tasks necessary to complete some project or reach some goal.

Counter intuitive: all producers can benefit, even when this means that one producer specializes in a good that another could produce at lower cost.

	Production if 100% of time is spent on one good
Greta	1,250 apples or 50 tons of wheat
Carlos	1,000 apples or 20 tons of wheat

Comparative advantage: when a person or country can produce a good at a *lower opportunity cost* than another person or country.

	Opportunity cost of 1 ton of wheat	Opportunity cost of 1 apple
Greta	1,250/50 = 25 apples	50/1,250 = 0.04 tons of wheat
Carlos	1,000/20 = 50 apples	20/1,000 = 0.02 tons of wheat

$\Rightarrow$ Carlos has a comparative advantage in apples because his relative cost of producing apples (that is, relative to wheat) is lower than Greta's.

Comparative advantage: Gains from specialization

If Carlos and Greta cannot trade with each other, they must each be self-sufficient, consuming exactly what they produce.

- Suppose that Greta chooses to use 40% of her time in apple production, and the rest producing wheat.
- Similarly Carlos spends 30% of his time producing apples, and 70% producing wheat.

		Self-sufficiency	**Specialization and trade**		
			Production	Consumption	Trade
		(1)	(2)	(3)	(4)
Greta	Apples	500			
	Wheat	30			
Carlos	Apples	300			
	Wheat	14			
Total	Apples	800			
	Wheat	44			

Comparative advantage: Gains from specialization

If they were to specialize, each producing only the crop in which they have a comparative advantage:

- Greta would produce 50 tons of wheat
- Carlos would produce 1,000 apples

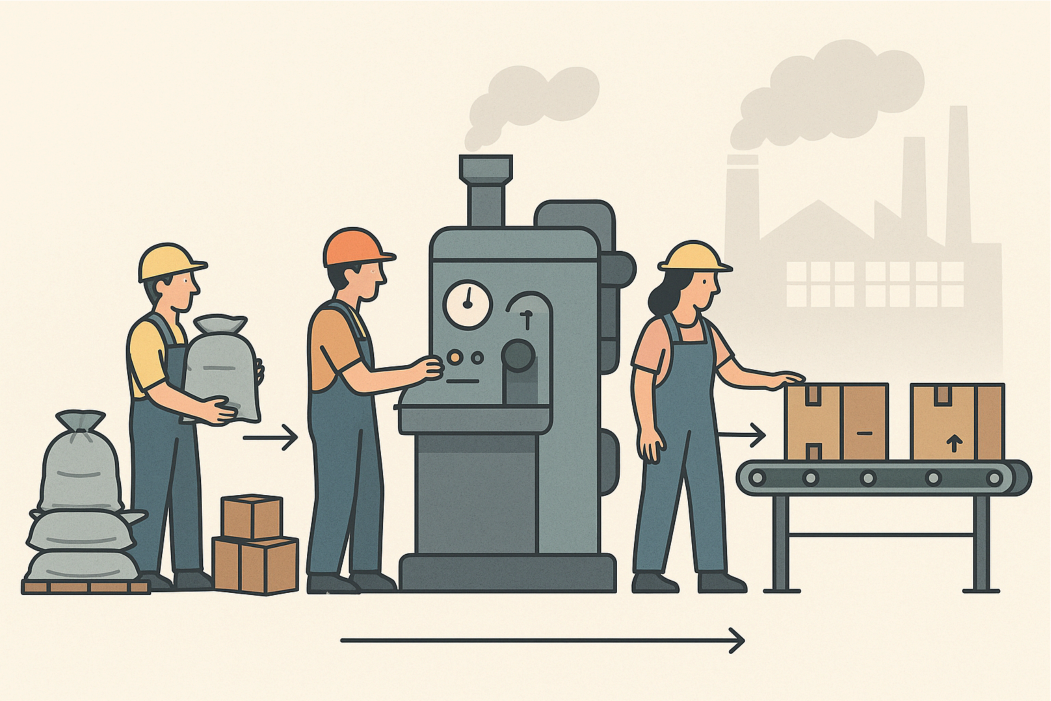
Suppose that they reach an agreement that both will specialize, and then Greta will sell 15 tons of wheat to Carlos in return for 600 apples.

		Self-sufficiency	**Specialization and trade**		
			Production	Consumption	Trade
		(1)	(2)	(3)	(4)
Greta	Apples	500	0	600	Buys 600 apples
	Wheat	30	50	35	Sells 15t wheat
Carlos	Apples	300	1,000	400	Sells 600 apples
	Wheat	14	0	15	Buys 15t wheat
Total	Apples	800	1,000	1,000	
	Wheat	44	50	50	

Firms, technology, and production

Firms, technology, and production

Technology: The description of a process using a set of materials and other inputs, including the work of people and machines, to produce an output.

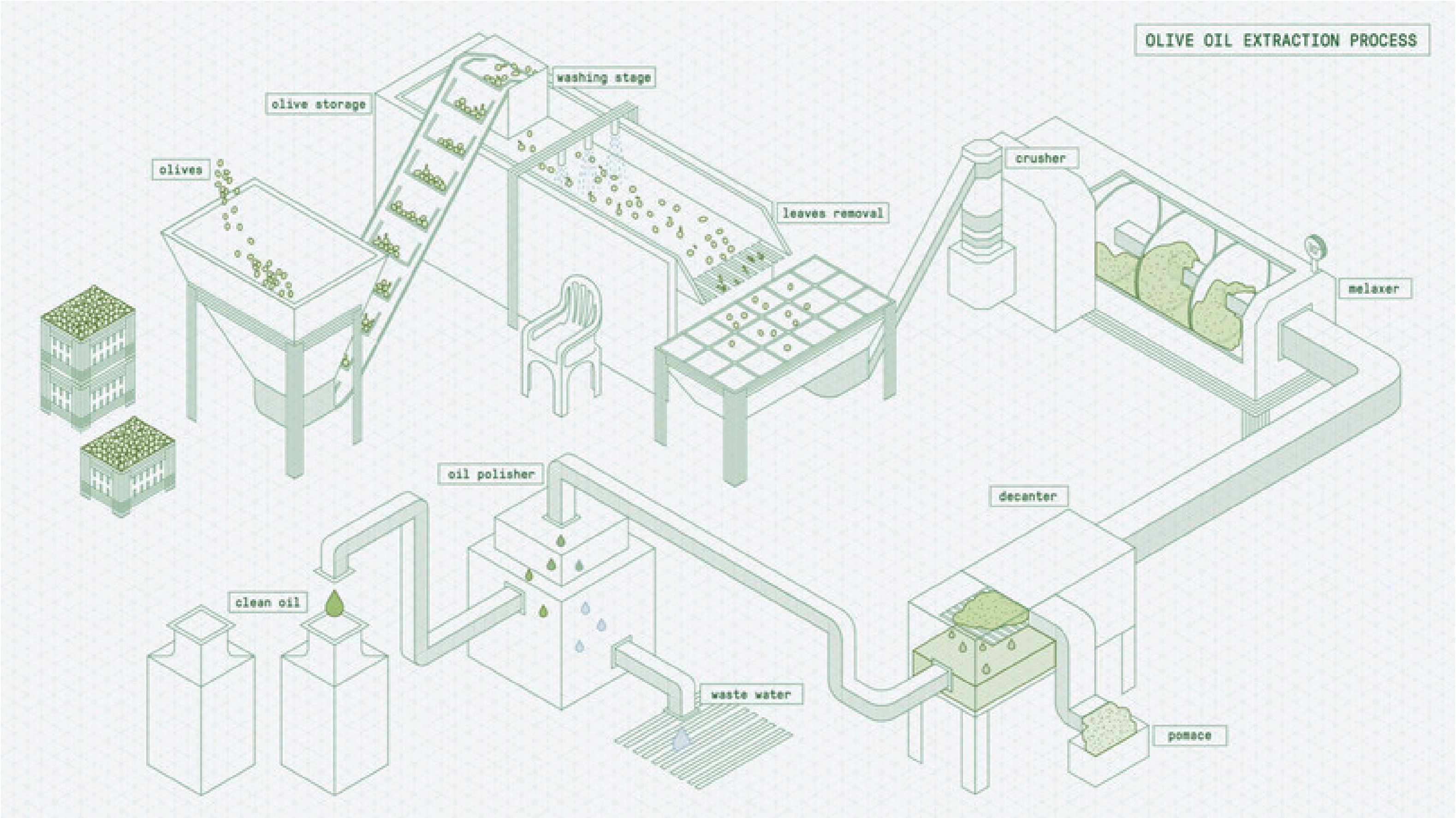


Having chosen a technology, need to decide on the amount of inputs to employ, which will determine how much output it can produce.

Firms, technology, and production

A labour-intensive technology for producing olive oil.

Firms, technology, and production



A capital-intensive olive technology.

Firms, technology, and production

A **technology can be represented by a production function**: a relationship that tells us how much output it will produce, given the amounts of inputs used.

$$Y = f(M, N, E)$$

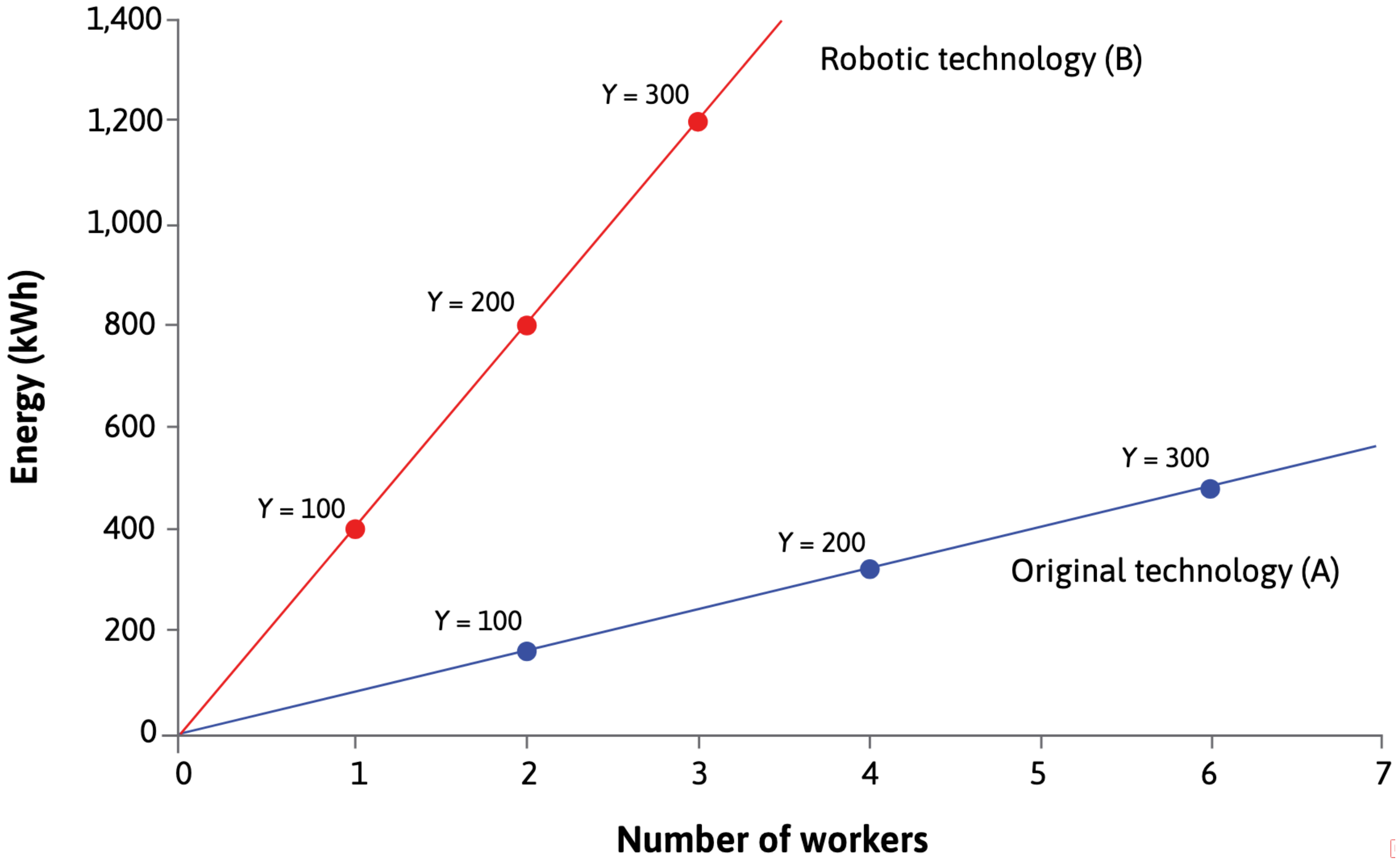
N number of workers employed, M number of machines, and E amount of energy used per day.

A hypothetical technology for olive oil:

Number of machines, *M*	**Number of workers, *N***	**Amount of energy, *E* (kWh)**	**Output, *Y* (litres)**
3	1	80	50
6	2	160	100
9	3	240	150
12	4	320	200

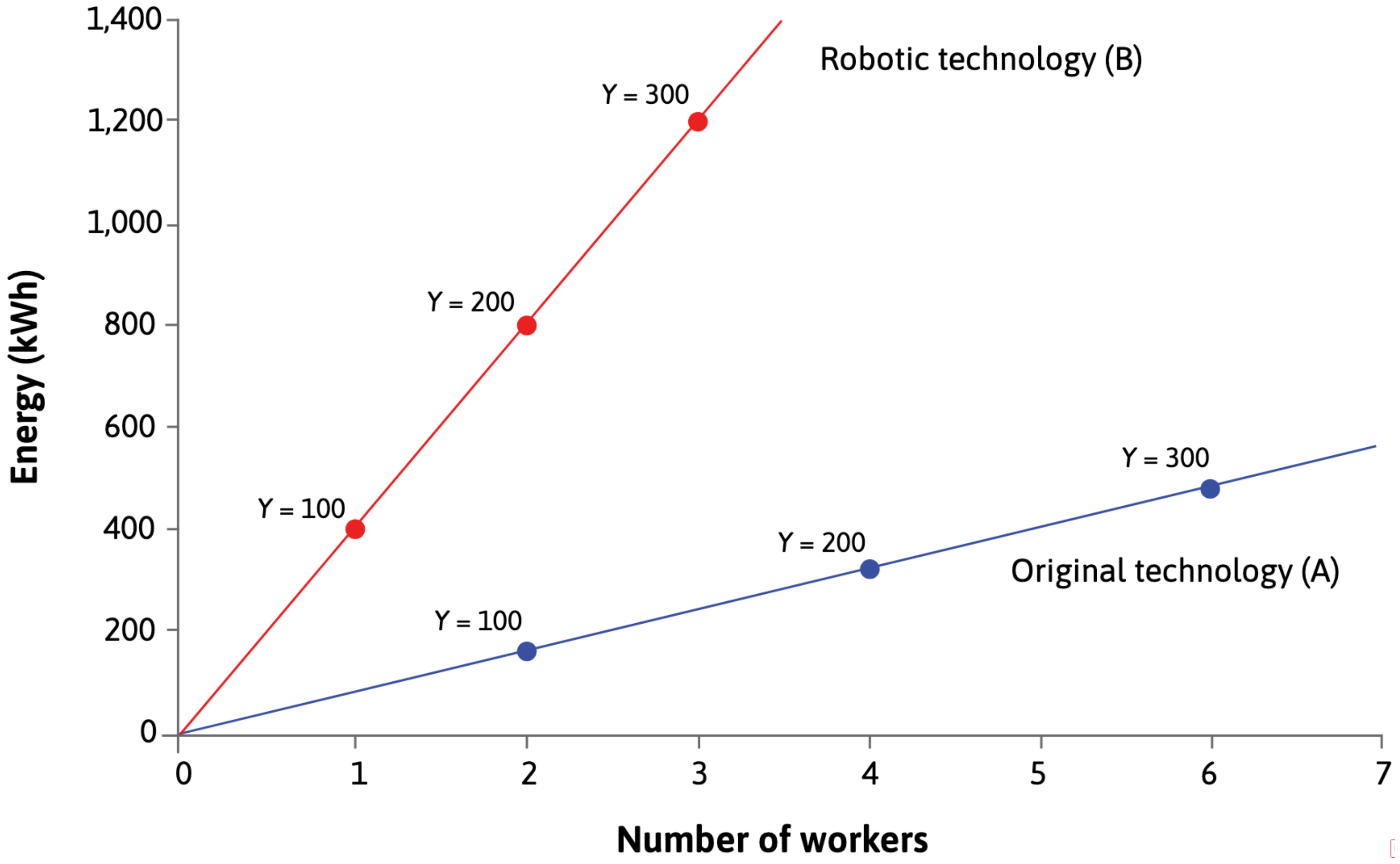
1. **fixed-proportions technology**: for every three machines, one worker, and 80 kWh of energy are needed.
2. **constant returns to scale**:if you double (or halve) the inputs, the amount of output doubles (or halves) accordingly.

Comparing two technologies



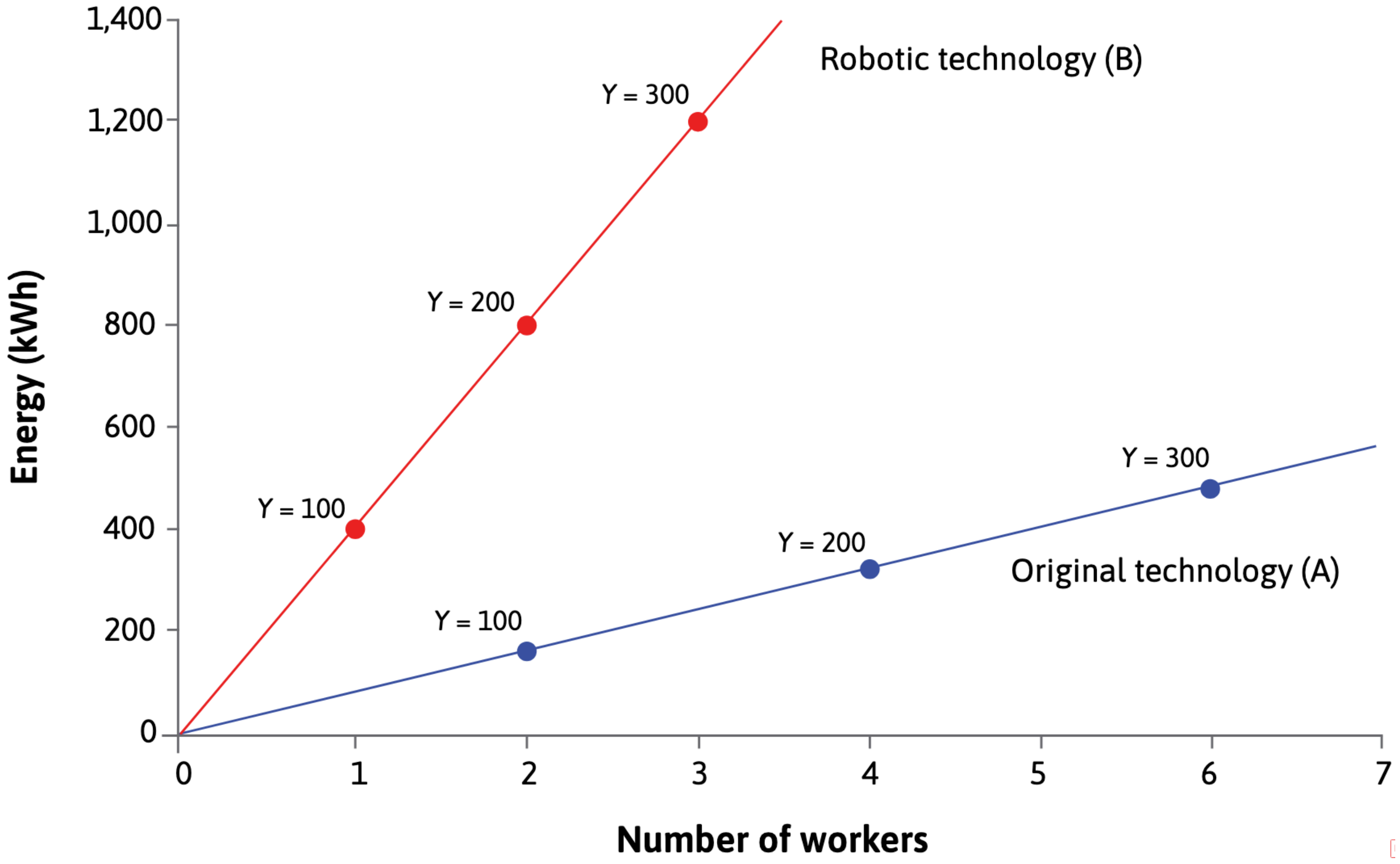
$$f_A(N = 2, E = 160) = 100$$

Comparing two technologies



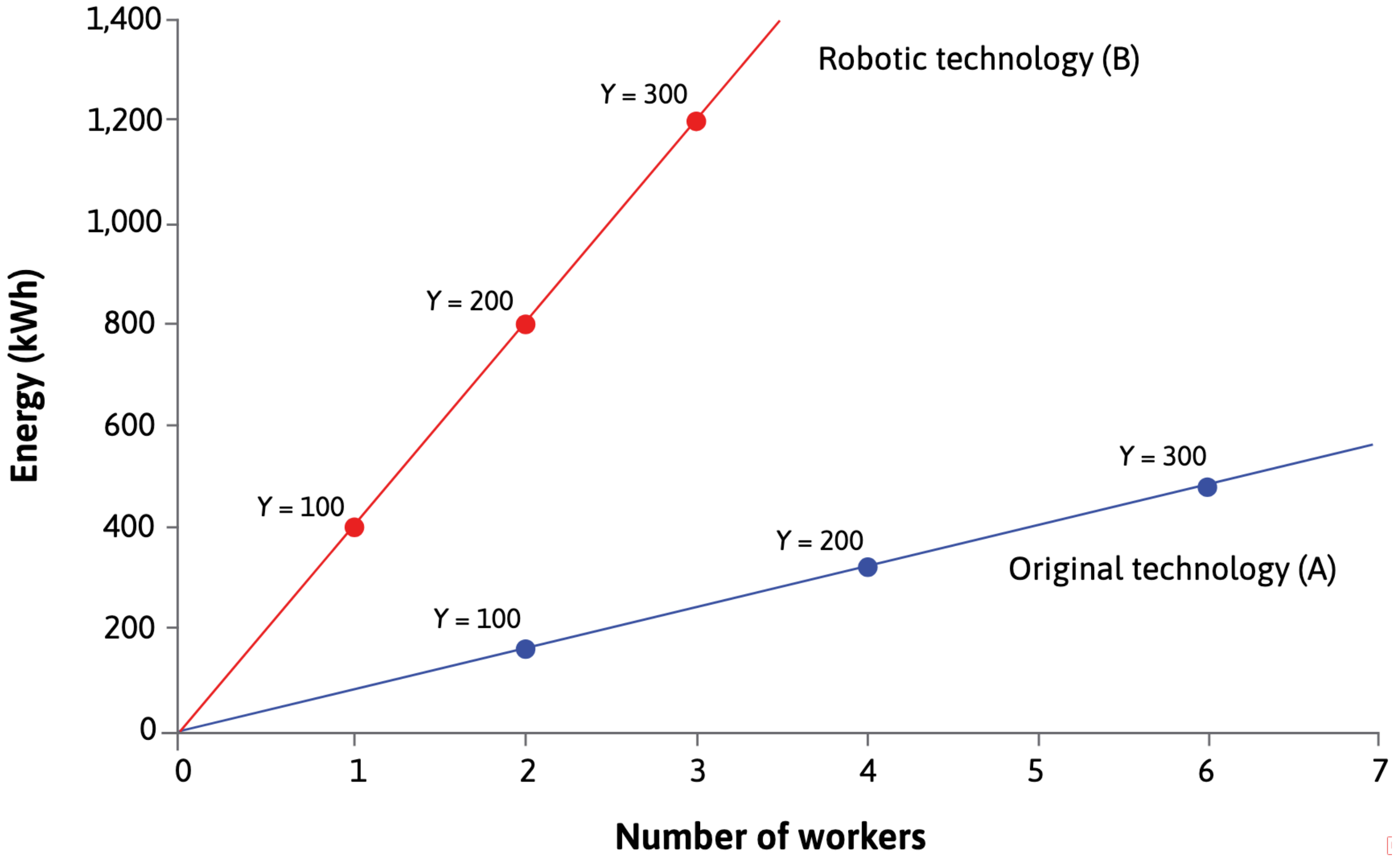
$$f_A(N = 2, E = 160) = 100 \quad \Rightarrow \textit{Average product of labour}_A = f_A(N = 2, E = 160)/2 = 100/2 = 50$$

Comparing two technologies



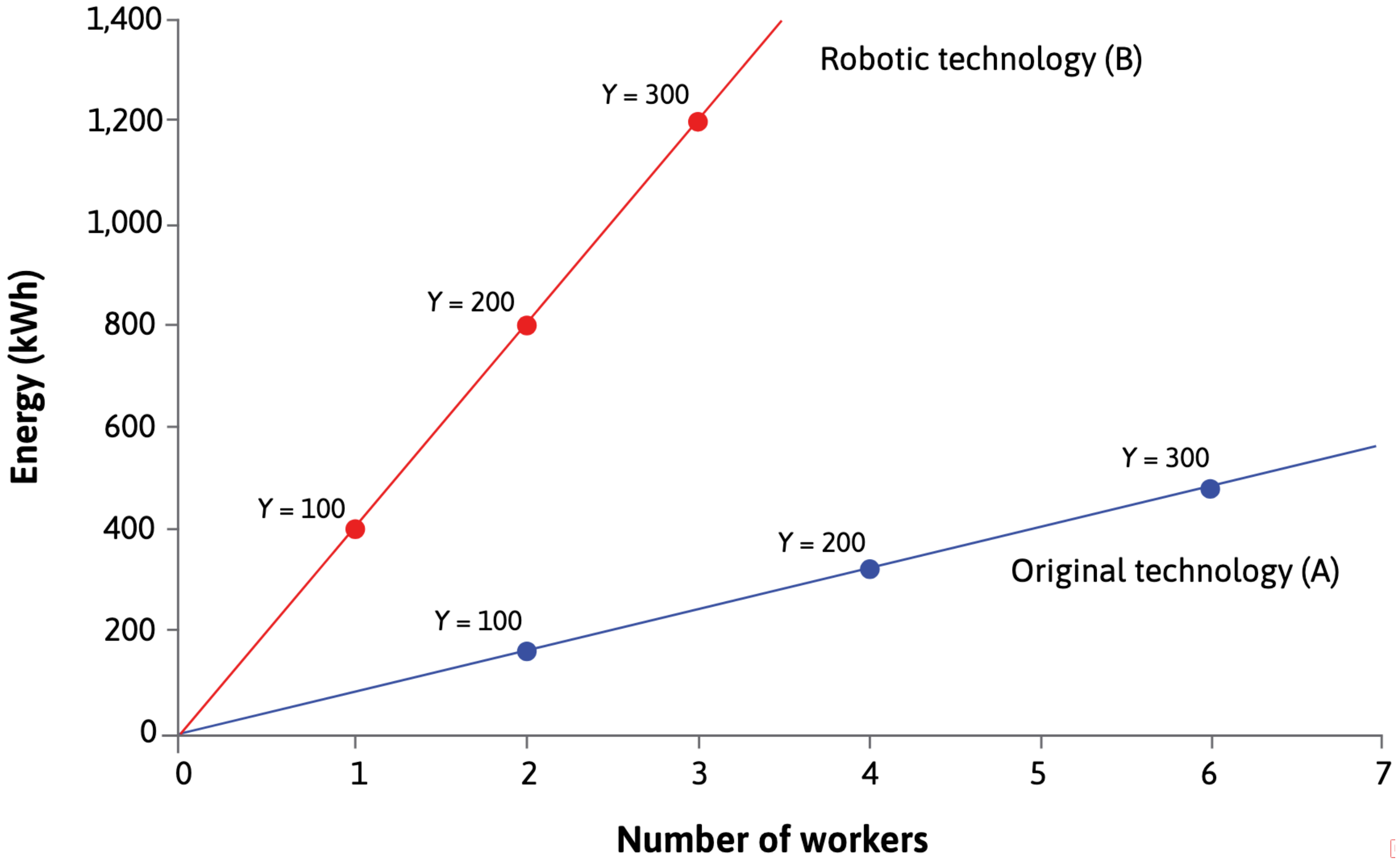
$$f_A(N = 2, E = 160) = 100 \quad \Rightarrow \textit{Energy–labour ratio}_A = 160/2 = 80$$

Comparing two technologies



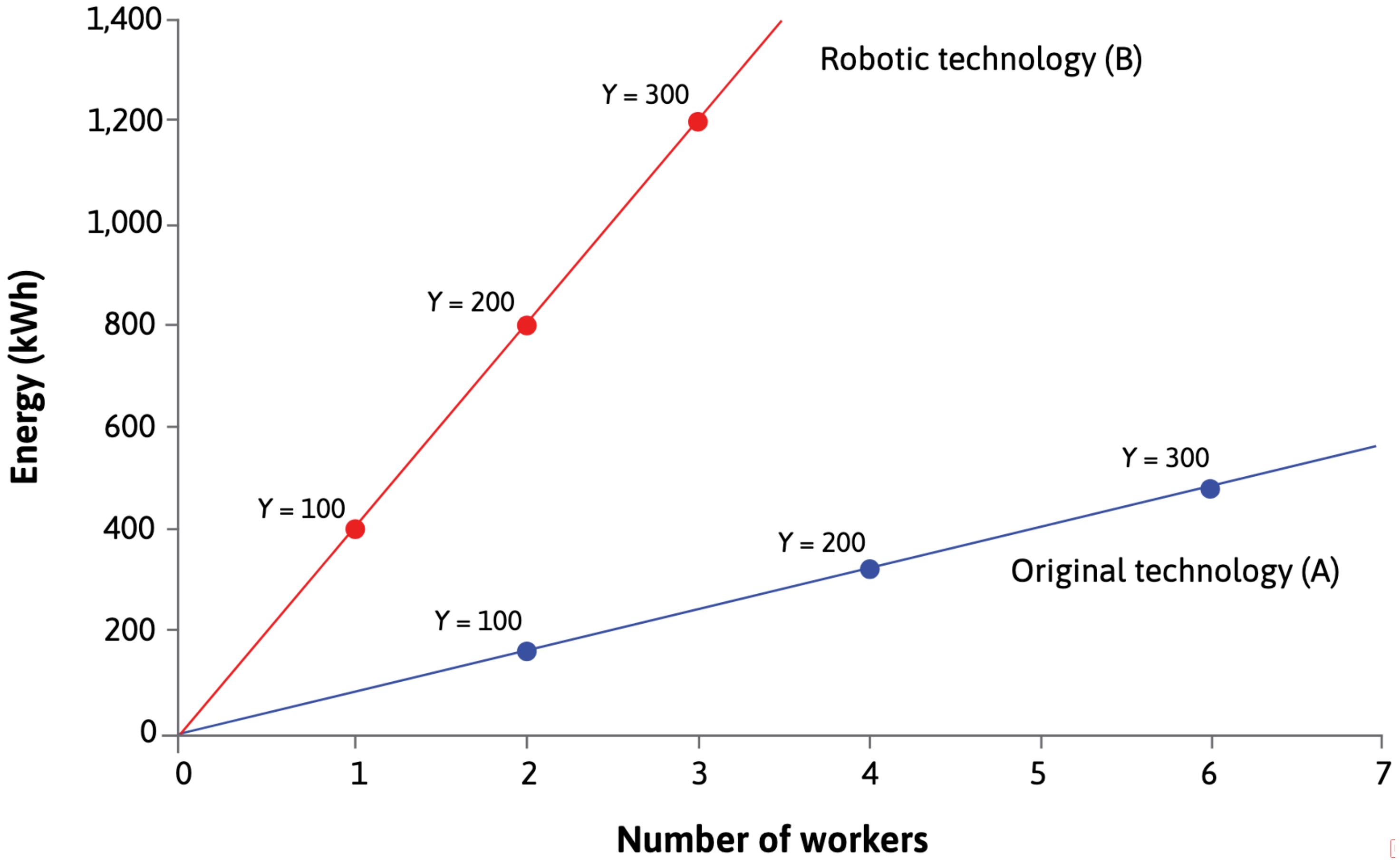
$$f_B(N = 1, E = 400) = 100$$

Comparing two technologies



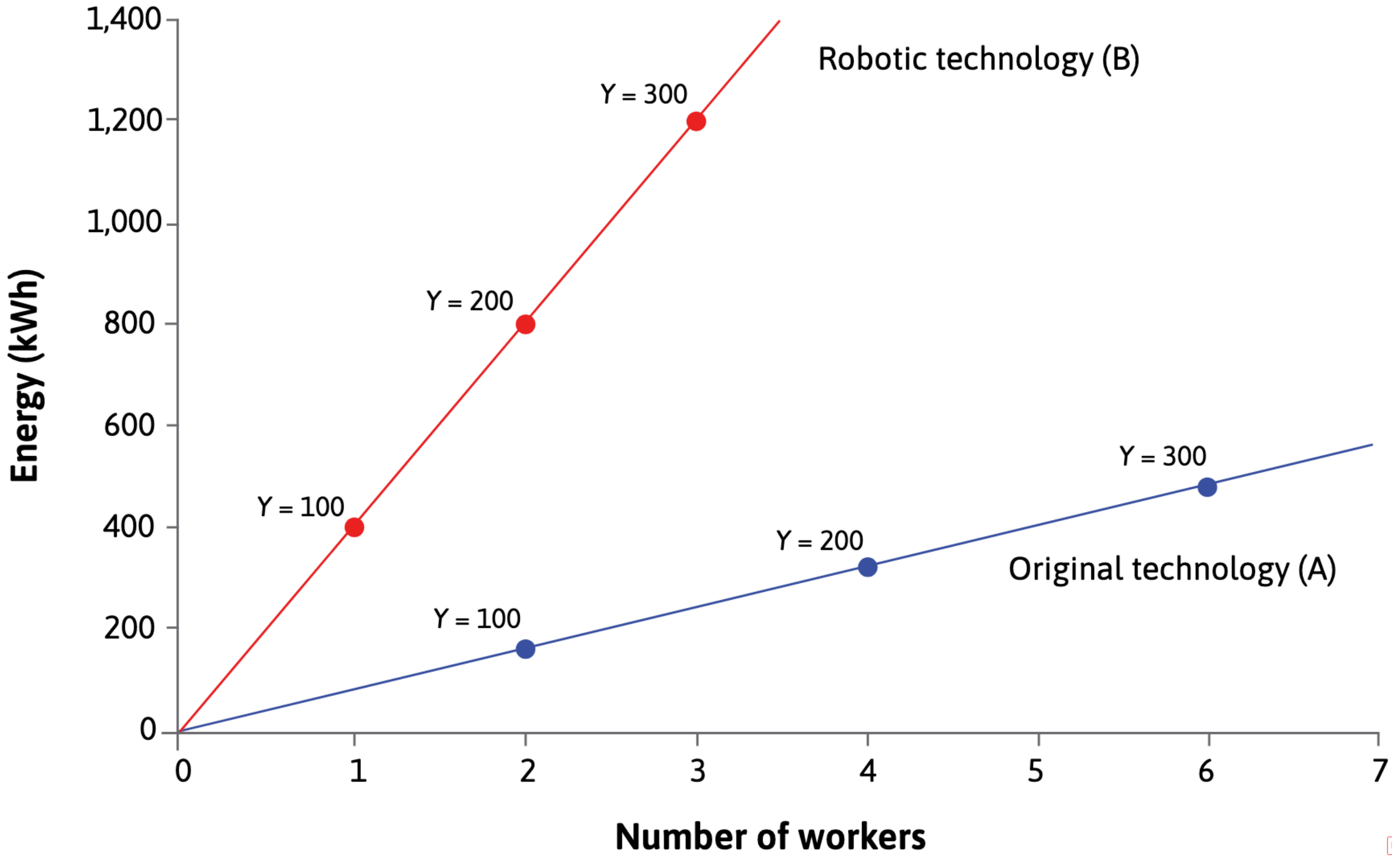
$$f_B(N=1, E=400) = 100 \quad \Rightarrow Average\ product\ of\ labour_B = f_B(N=1, E=400)/1 = 100/1 = 100$$

Comparing two technologies



$$f_B(N = 1, E = 400) = 100 \quad \Rightarrow \textit{Energy–labour ratio}_B = 400/1 = 400$$

Comparing two technologies



Which technology should be used?

Modelling a dynamic economy: Technology and costs

Modelling a dynamic economy: Technology and costs

Five different technologies to produce 100 metres of cloth.

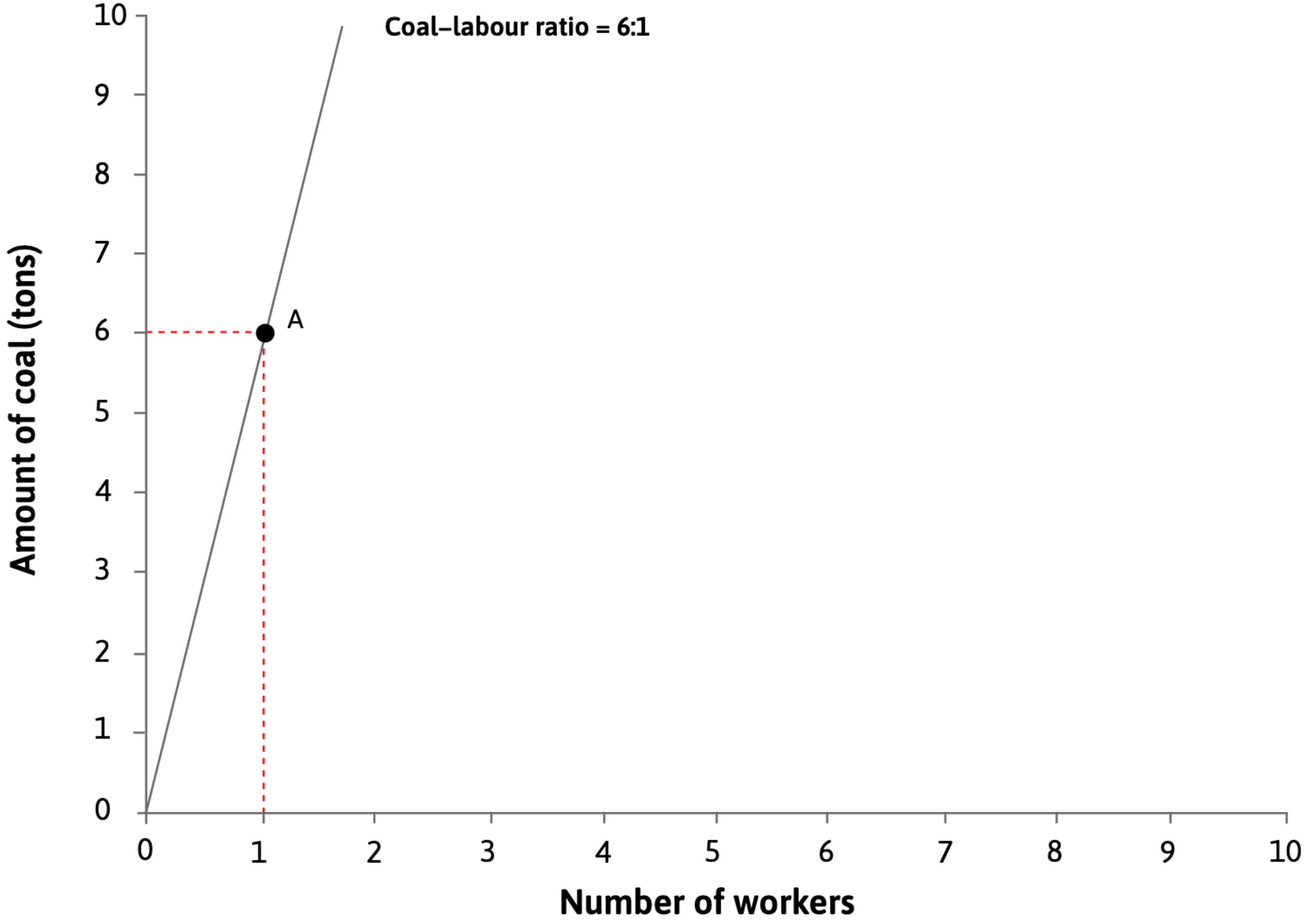
Inputs: labour (number of workers) and energy (tons of coal).

Technology	Number of workers	Coal required (tons)
A	1	6
B	4	2
C	3	7
D	5	5
E	10	1

Modelling a dynamic economy: Technology and costs

Tech.	No. of workers	Coal (tons)
A	1	6
B	4	2
C	3	7
D	5	5
E	10	1

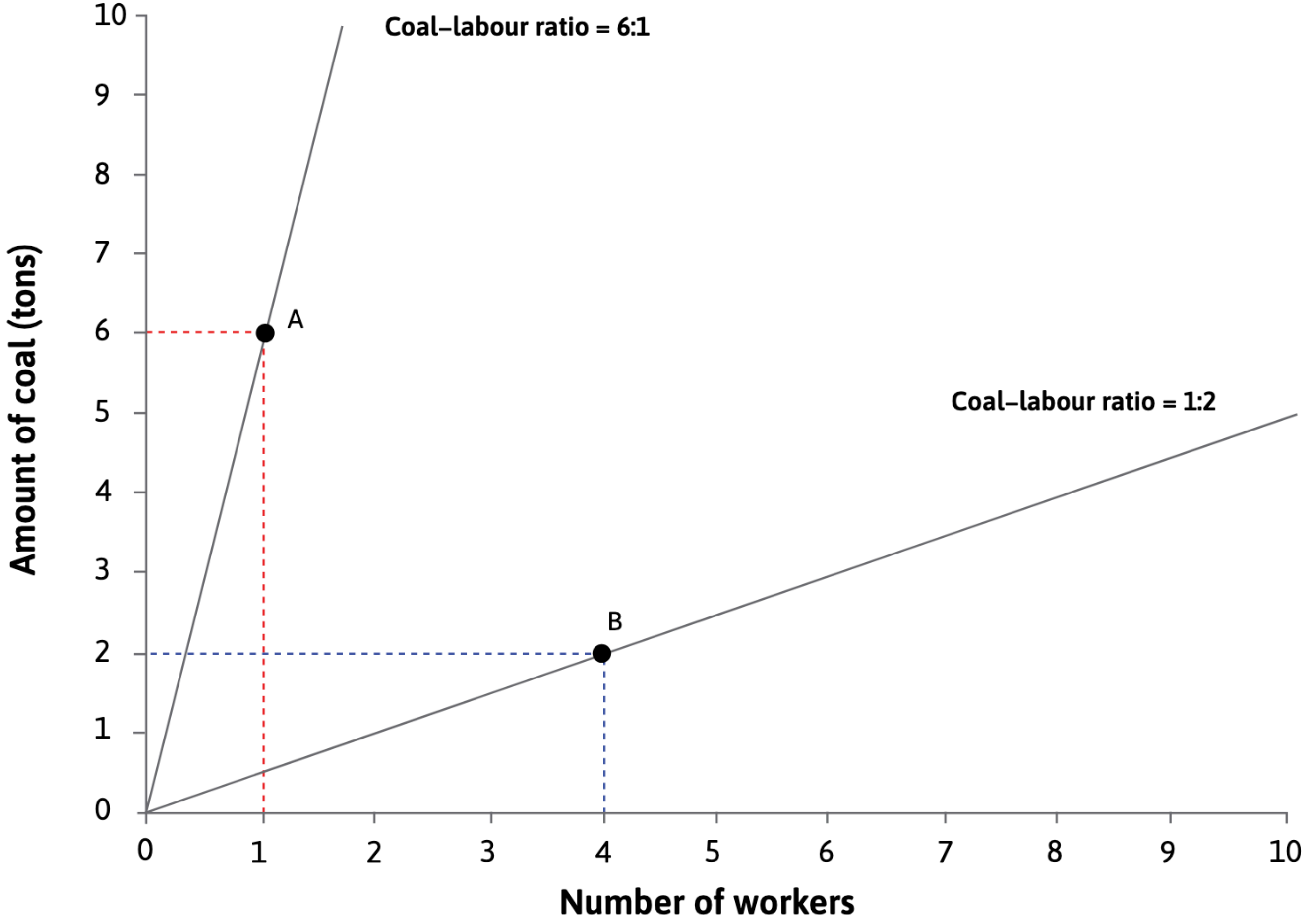
Technology A is the most energy-intensive, using one worker and six tons of coal.



Modelling a dynamic economy: Technology and costs

Tech.	No. of workers	Coal (tons)
A	1	6
B	4	2
C	3	7
D	5	5
E	10	1

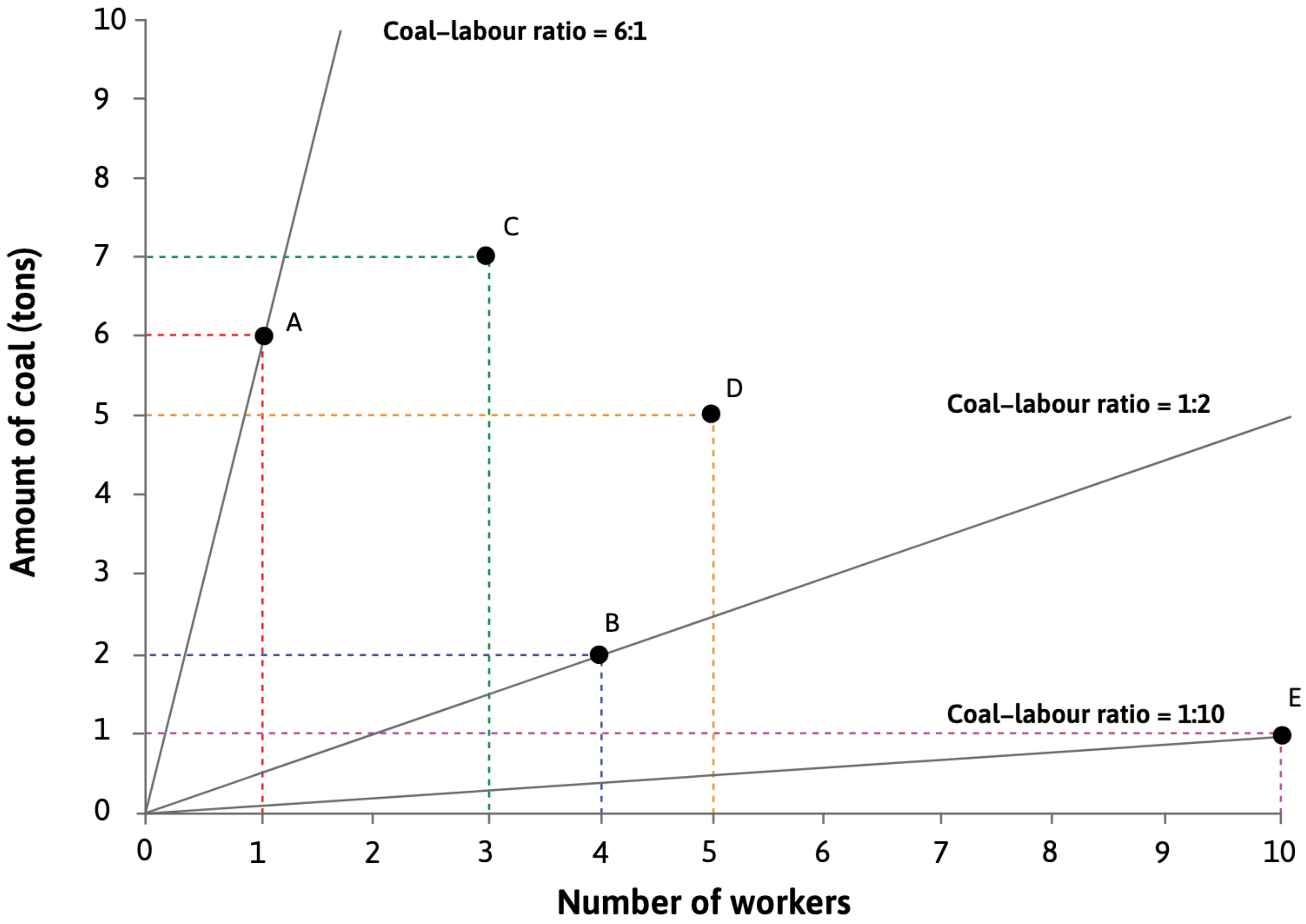
Technology B : it is more labour-intensive than technology A, because the ratio of workers to coal required to produce the given output is higher for B than A.



Modelling a dynamic economy: Technology and costs

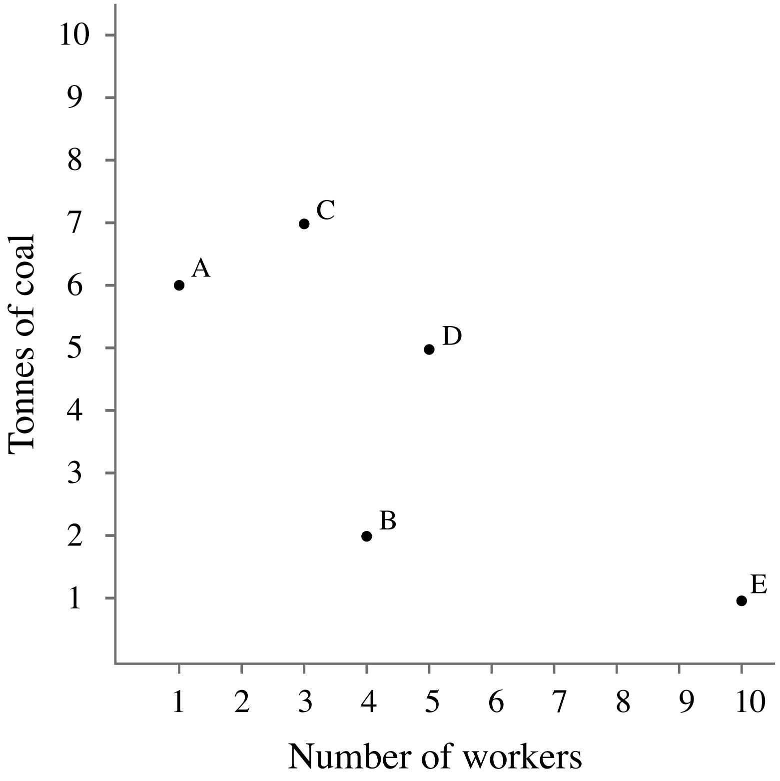
Tech.	No. of workers	Coal (tons)
A	1	6
B	4	2
C	3	7
D	5	5
E	10	1

Technology E is the most labour-intensive of the five technologies: it uses ten workers and one ton of coal.

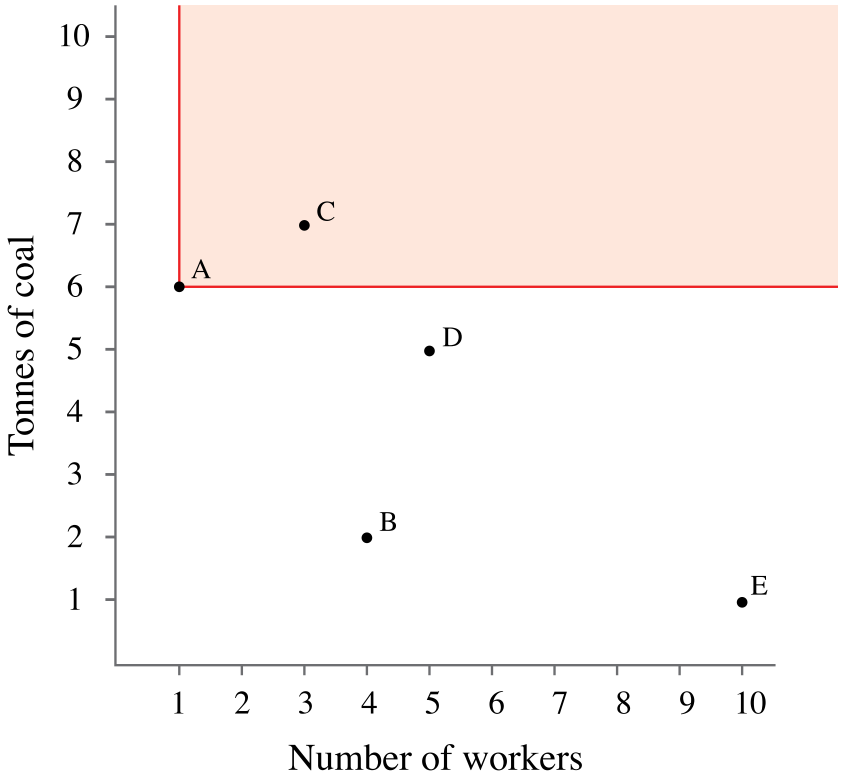


Modelling a dynamic economy: Technology and costs

Which technology would the firm choose?

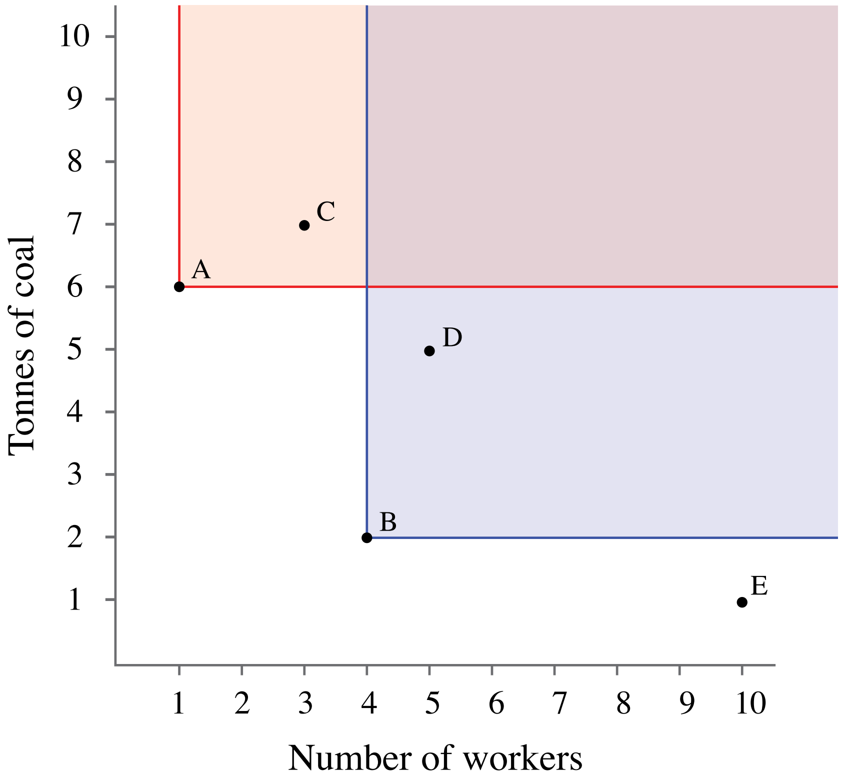


Modelling a dynamic economy: Technology and costs



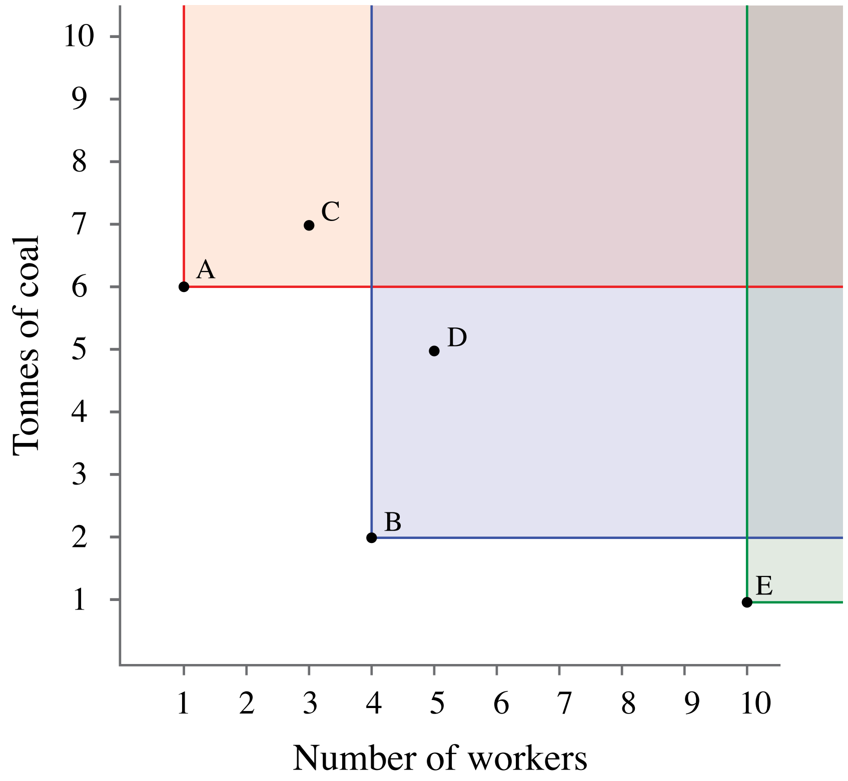
Technology A dominates the C-technology: the same amount of cloth can be produced using A with fewer inputs of labour and energy. This means that, whenever A is available, you would never use C.

Modelling a dynamic economy: Technology and costs



Technology B dominates the D-technology: the same amount of cloth can be produced using B with fewer inputs of labour and energy. Note that B would dominate any other technology that is in the shaded area above and to the right of point B.

Modelling a dynamic economy: Technology and costs



Technology E does not dominate any of the available technologies. None of them are in the area above and to the right of E. Technology E is also not dominated by any other available technology.

How does a firm evaluate the cost of production using different technologies?

Firms aim to maximize their profit, which means producing cloth at the least possible cost.

This is why the firms' choice of technology depends on economic information about relative prices of inputs.

$$cost = (wage \times workers) + (price\ of\ a\ tonne\ of\ coal \times\ number\ of\ tonnes\ of\ coal)$$

$$c = (w \times L) + (p \times R)$$

$$c = wL + pR$$

Then

$$pR = c - wL$$

$$R = \frac{c}{p} - \frac{w}{p}L$$

R = **Isocost lines**: combinations of inputs that give the same cost (slope = relative price of inputs)

$-\frac{w}{p}$: slope $\rightarrow$ **relative price of labour**

Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow \textit{Relative price}: \frac{w}{p} = \frac{10}{20} = .5$

The total cost of employing 2 workers with 3 tonnes of coal is (2 × 10) + (3 × 20) = £80.

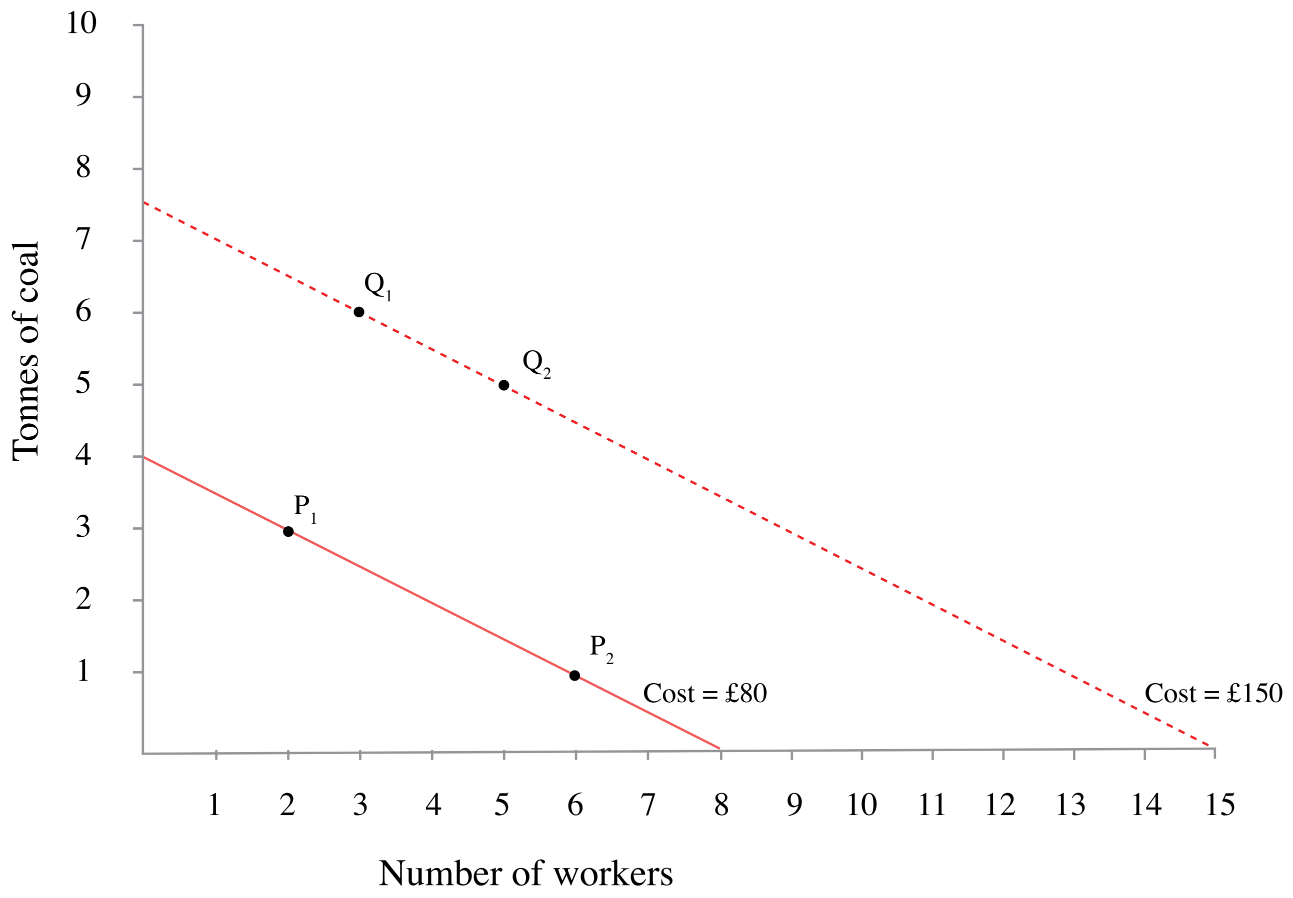
Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow Relative\ price: \frac{w}{p} = \frac{10}{20} = .5$

If the number of workers is increased to 6, costing £60, and the input of coal is reduced to 1 tonne, the total cost will still be £80.

Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow$ $Relative\ price: \frac{w}{p} = \frac{10}{20} = .5$

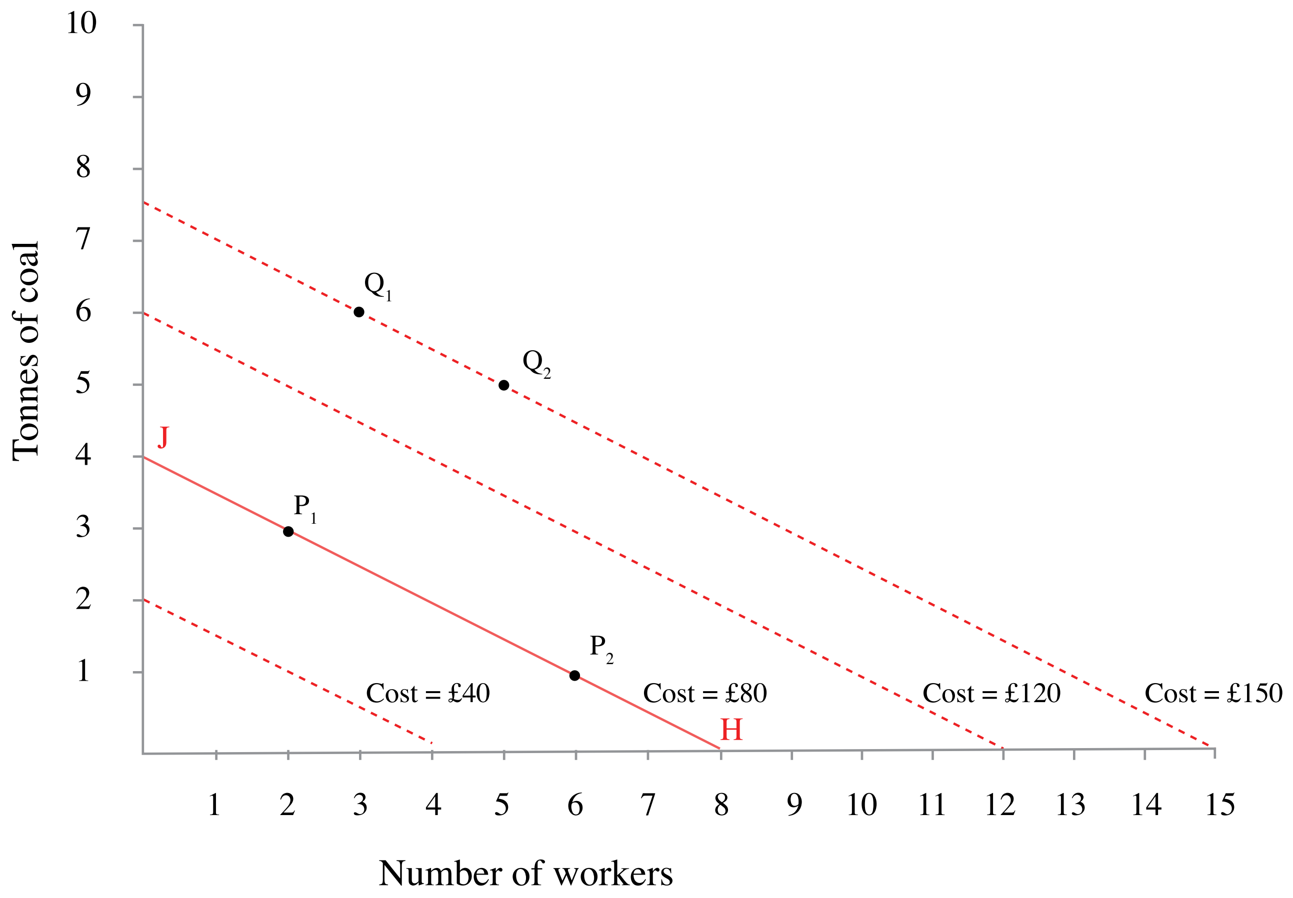
The straight line through P1 and P2 joins together all the points where the total cost is £80.

Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow$ $\mathit{Relative\ price}: \frac{w}{p} = \frac{10}{20} = .5$



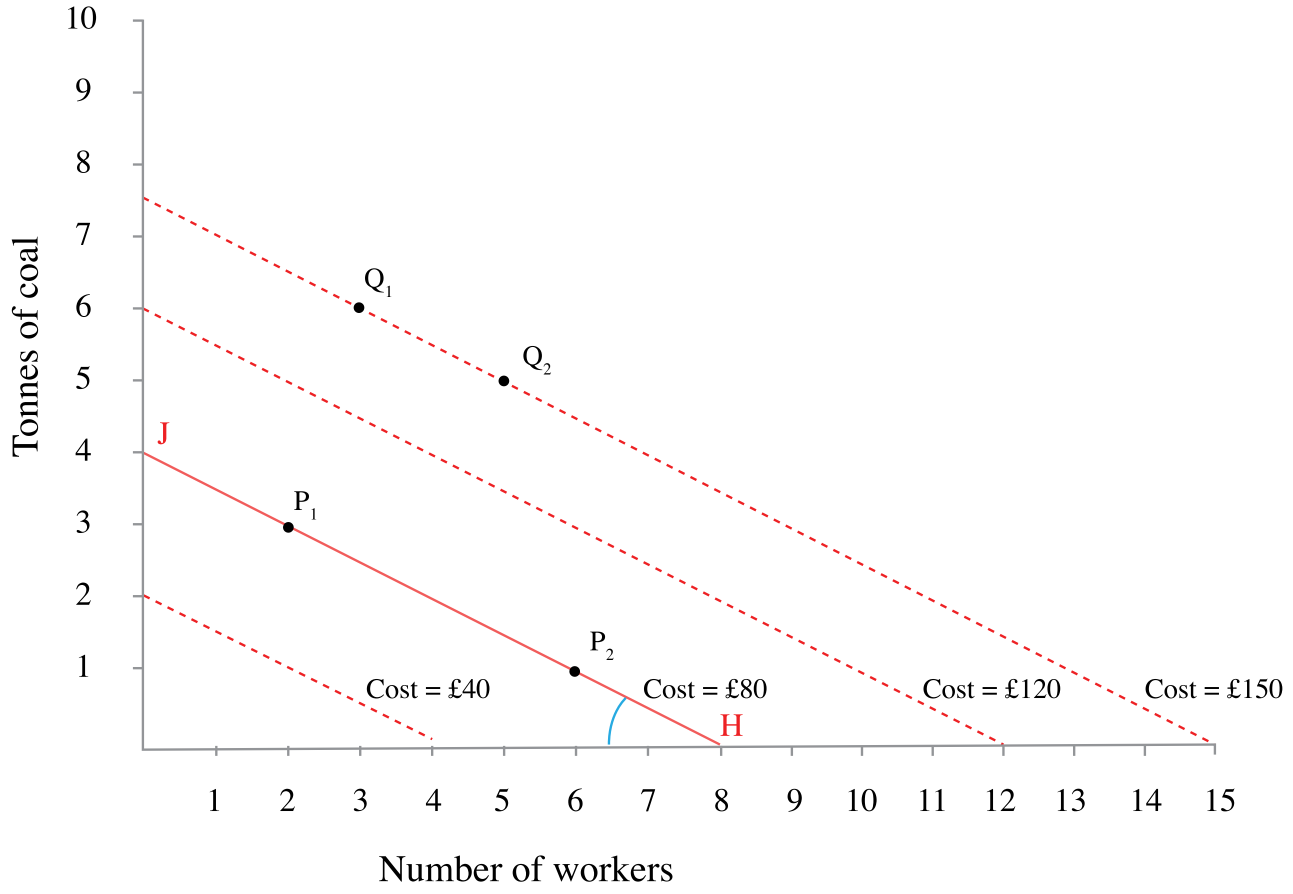
At point Q1 (3 workers, 6 tonnes of coal) the total cost is £150. In point Q2, if 2 more workers are employed, the input of coal should be reduced by 1 tonne to keep the cost at £150. This is the £150 isocost line.

Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow$ $Relative\ price: \frac{w}{p} = \frac{10}{20} = .5$



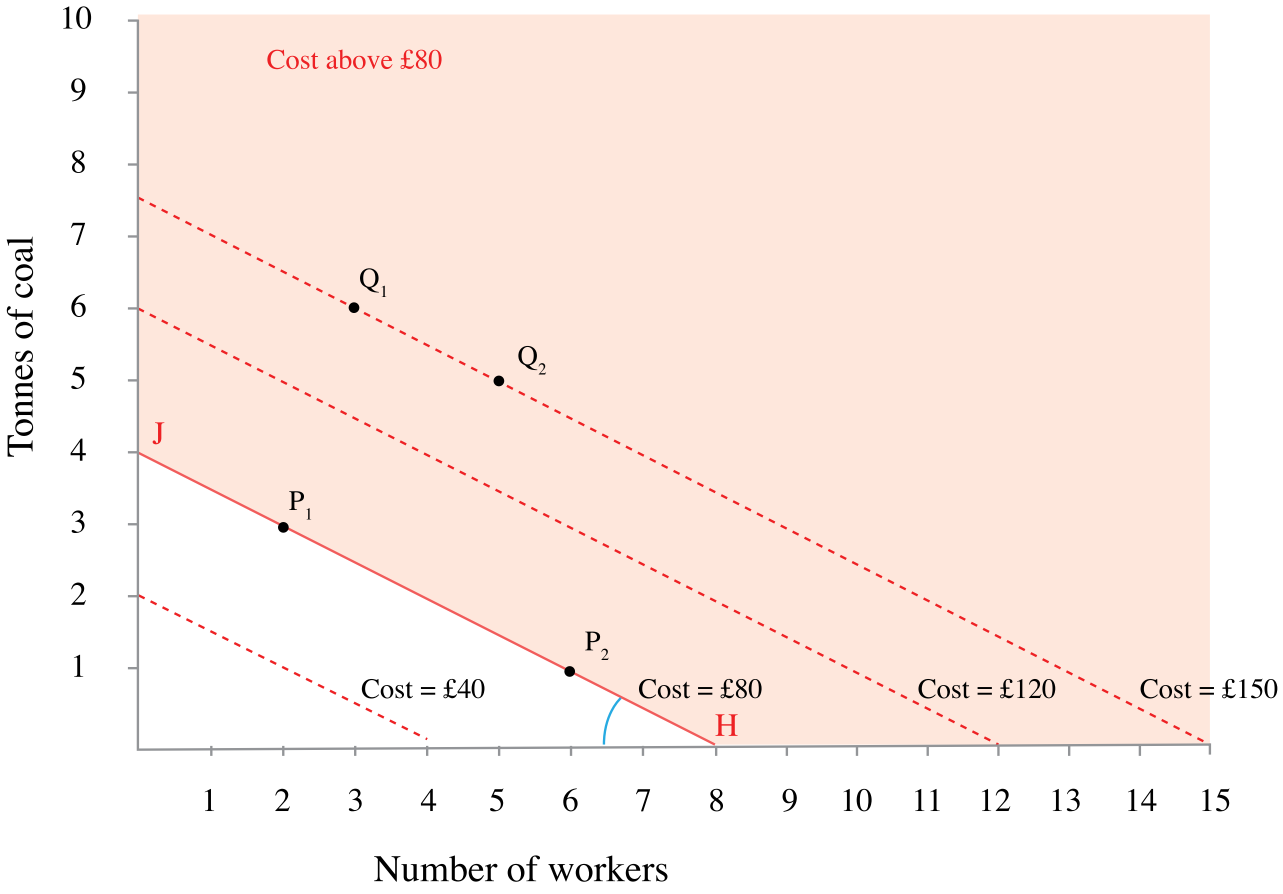
If prices of inputs are fixed, the isocost lines are parallel. A simple way to draw any line is to find the end points: for example, the £80 line joins the points J (4 tonnes of coal and no workers) and H (8 workers, no coal).

Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow$ $Relative\ price : \frac{w}{p} = \frac{10}{20} = .5$



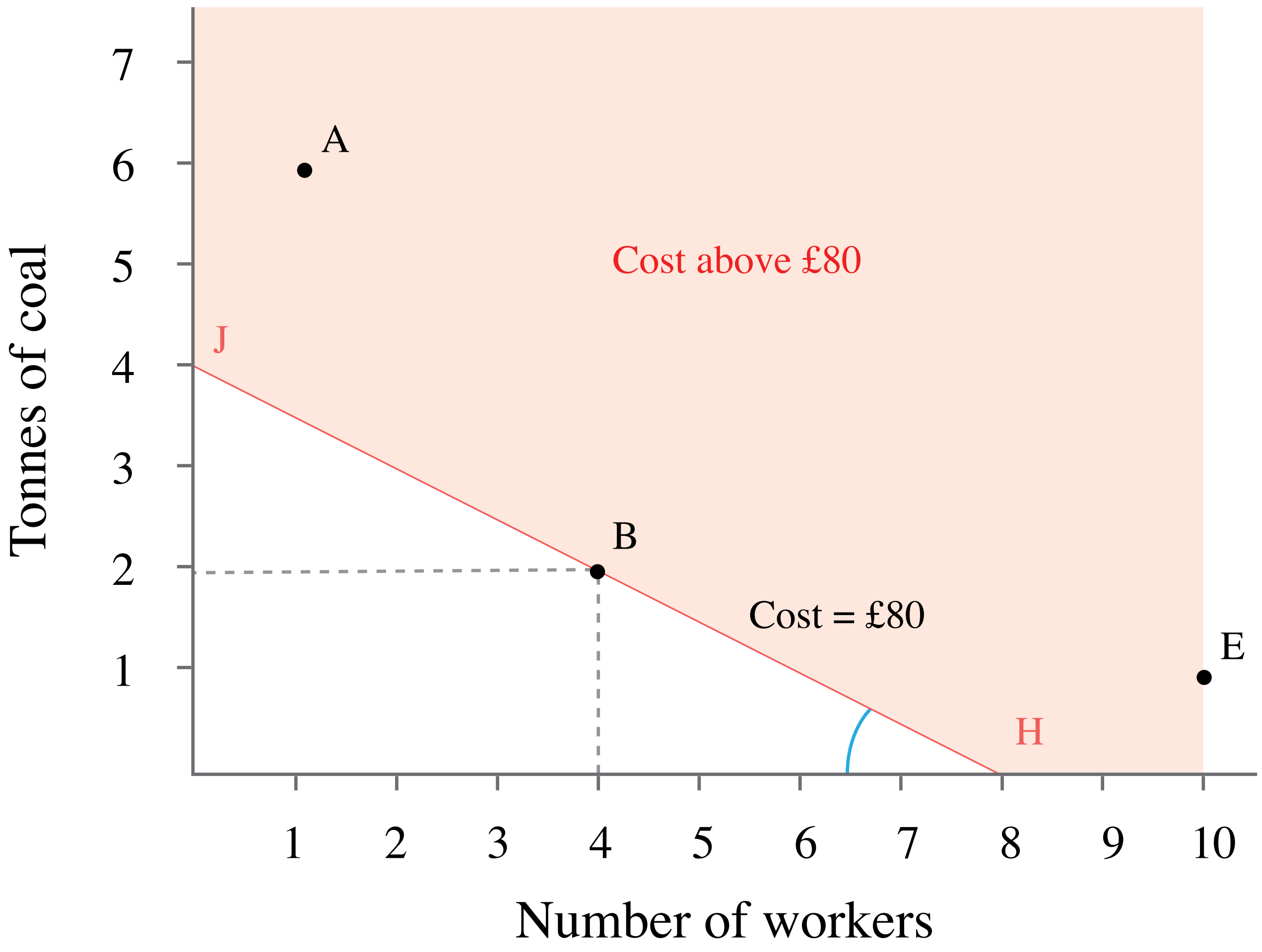
The slope of the isocost lines is negative. In this case the slope is –0.5, because at each point, if you hired one more worker, costing £10, and reduced the amount of coal by 0.5 tonnes, saving £10.

Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow$ $Relative\ price: \frac{w}{p} = \frac{10}{20} = .5$



Points above an isocost line cost more.

Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow$ $\textit{Relative price}: \frac{w}{p} = \frac{10}{20} = .5$



Which is the least-cost technology? **B**

Suppose that the wage is £10 and the price of coal is £20 per tonne $\rightarrow \textit{Relative price}: \frac{w}{p} = \frac{10}{20} = .5$

Technology	Number of workers	Coal required (tonnes)	Total cost (£)
B	4	2	80
A	1	6	130
E	10	1	120

Wage £10, cost of coal £20 per tonne

Which is the least-cost technology? **B**

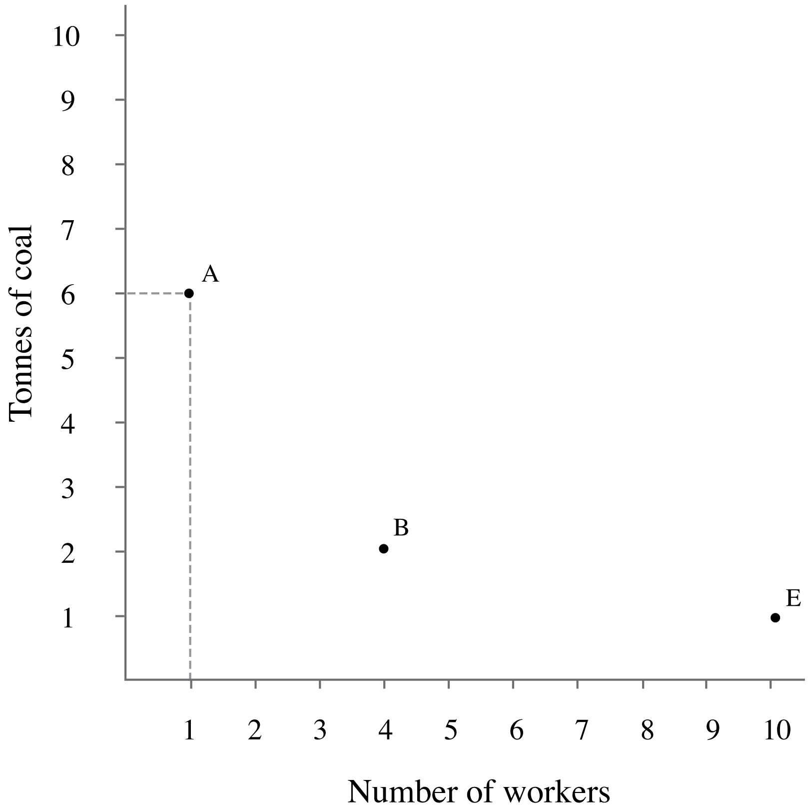
Modelling a dynamic economy: Innovation and profit

Modelling a dynamic economy: Innovation and profit

Suppose that the price of coal falls to £5 while the wage remains at £10 $\rightarrow Relative\ price : \frac{w}{p} = \frac{10}{5} = 2$

Modelling a dynamic economy: Innovation and profit

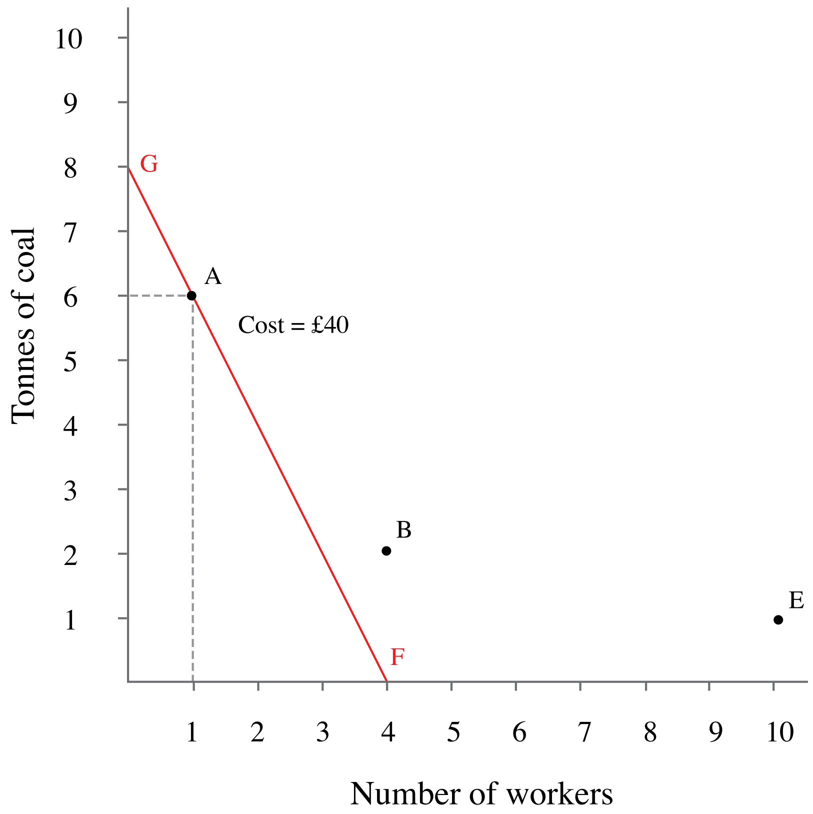
Suppose that the price of coal falls to £5 while the wage remains at £10 $\rightarrow$ $Relative\ price: \frac{w}{p} = \frac{10}{5} = 2$



A-technology, which is more energy-intensive than the others, can produce 100 metres of cloth at a lower cost than B or E.

Modelling a dynamic economy: Innovation and profit

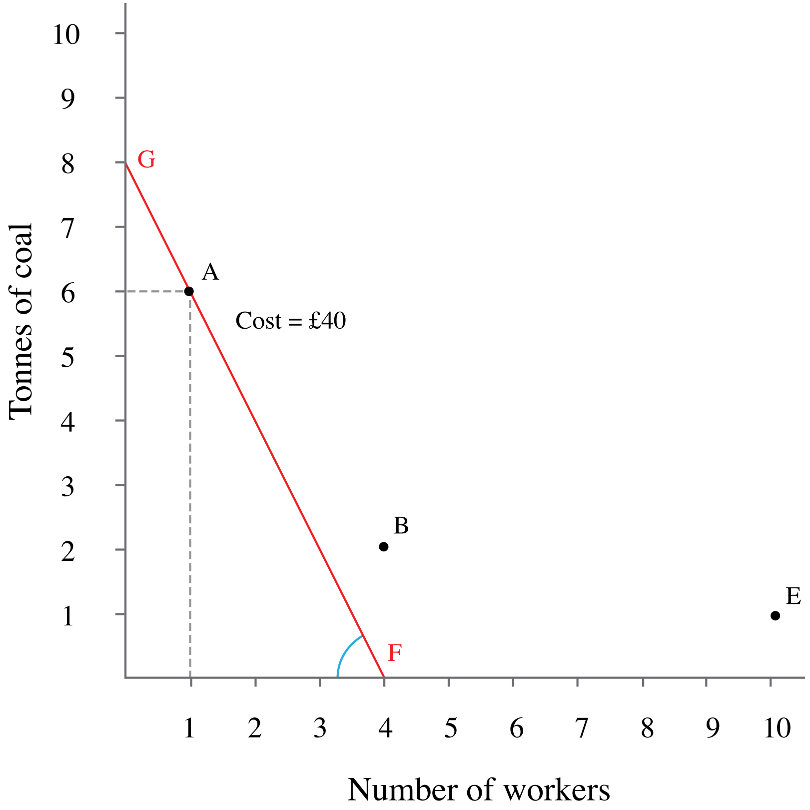
Suppose that the price of coal falls to £5 while the wage remains at £10 $\rightarrow$ $Relative\ price: \frac{w}{p} = \frac{10}{5} = 2$



The A-technology is on the isocost line FG. At any point on this line, the total cost of inputs is £40. Technologies B and E are above this line, with higher costs.

Modelling a dynamic economy: Innovation and profit

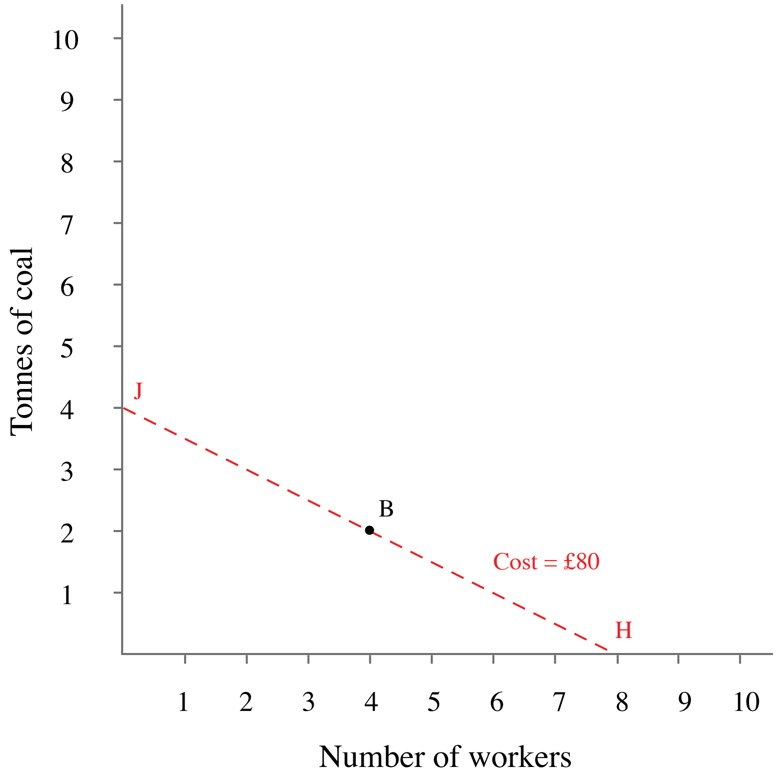
Suppose that the price of coal falls to £5 while the wage remains at £10 $\rightarrow$ $Relative\ price : \frac{w}{p} = \frac{10}{5} = 2$



The slope of the isocost line is equal to –(10/5) = –2. If you spent £10 on labour by hiring an extra worker, you could reduce coal by 2 tonnes and keep the total cost at £40.

Modelling a dynamic economy: Innovation and profit

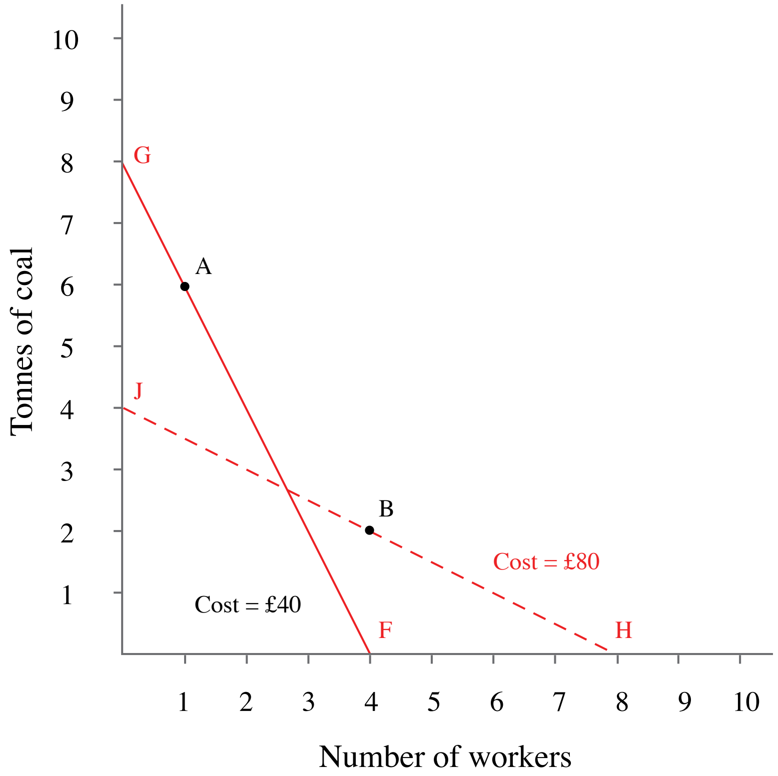
At the original relative price, B is the lower cost technology



When the wage is £10 and the price of coal is relatively high at £20, the cost of producing 100 metres of cloth using technology B is £80: choosing the B-technology puts the firm on the HJ isocost curve.

Modelling a dynamic economy: Innovation and profit

The price of coal falls to £5



If the price of coal falls relative to the wage as shown by the isocost curve FG, then using the A-technology, which is more energy-intensive than B, costs £40.

Modelling a dynamic economy: Innovation and profit

The price of coal falls to £5

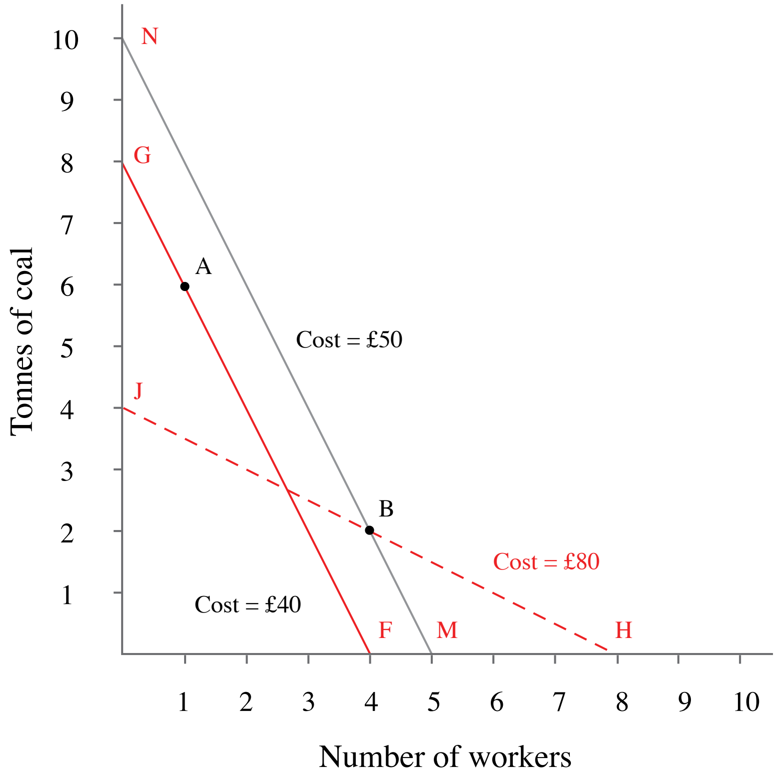
Technology	Number of workers	Coal required (tonnes)	Total cost (£)
B	4	2	50
A	1	6	40
E	10	1	105

Wage £10, cost of coal £5 per tonne

If the price of coal falls relative to the wage as shown by the isocost curve FG, then using the A-technology, which is more energy-intensive than B, costs £40.

Modelling a dynamic economy: Innovation and profit

B now costs more than A



At the new relative prices, the B-technology is on the isocost line MN, where the cost is £50. Switching to technology A will be cheaper.

Modelling a dynamic economy: Innovation and profit

$$profits = revenue - costs$$

Switching from A to B:

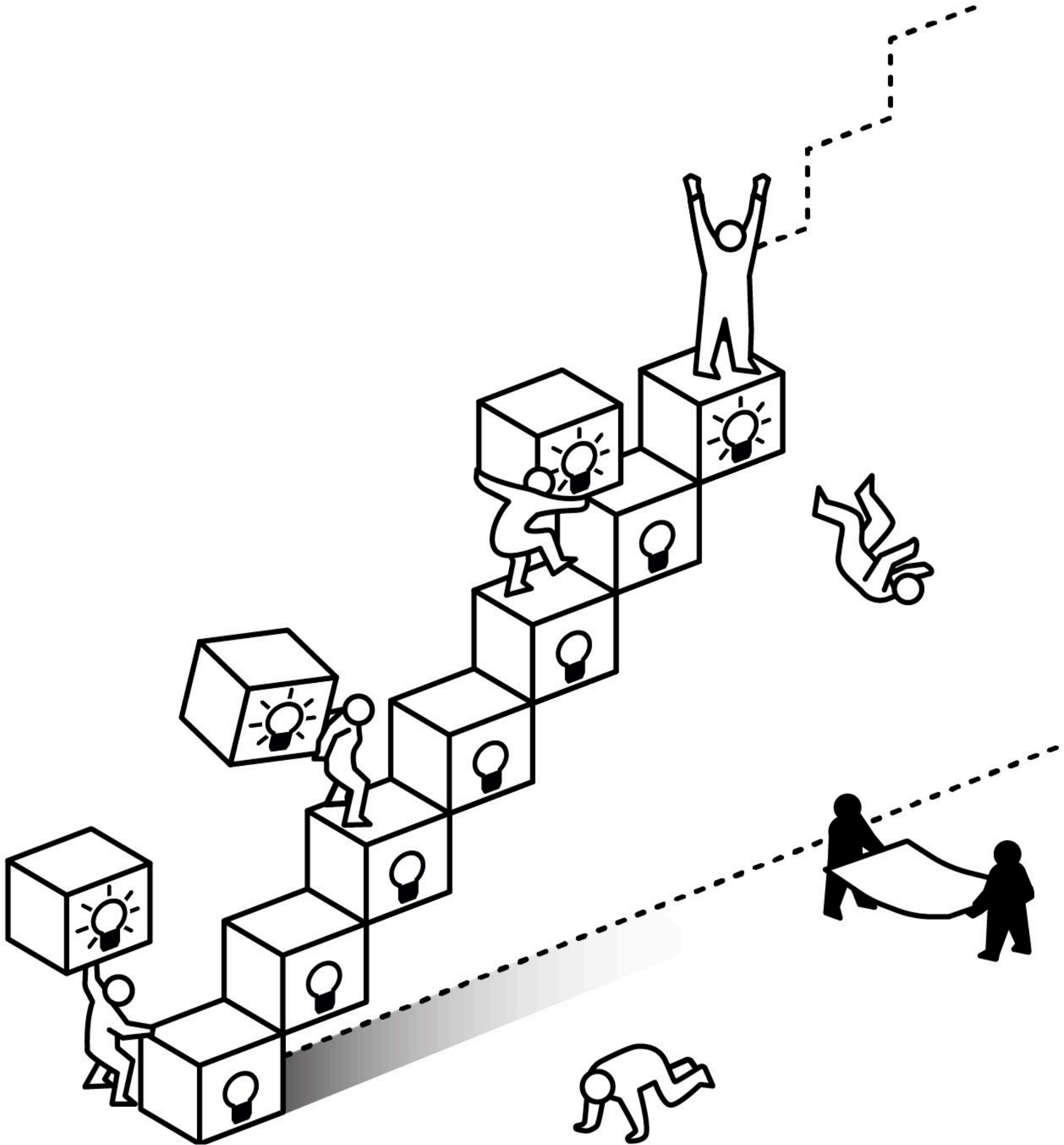
$$\Delta profits = \Delta revenue - \Delta costs = 0 - (40 - 50) = 10$$

The economic rent for a firm switching from B to A is £10 per 100 metres of cloth, which is the cost reduction made possible by the new technology

Schumpeterian (innovation) rents

Joseph Schumpeter: the adoption of technological improvements by **entrepreneurs** --first adopters-- a key part of his explanation for the dynamism of capitalism.

Creative Destruction: the process by which old technologies and the firms that do not adapt are swept away by the new, because they cannot compete in the market.

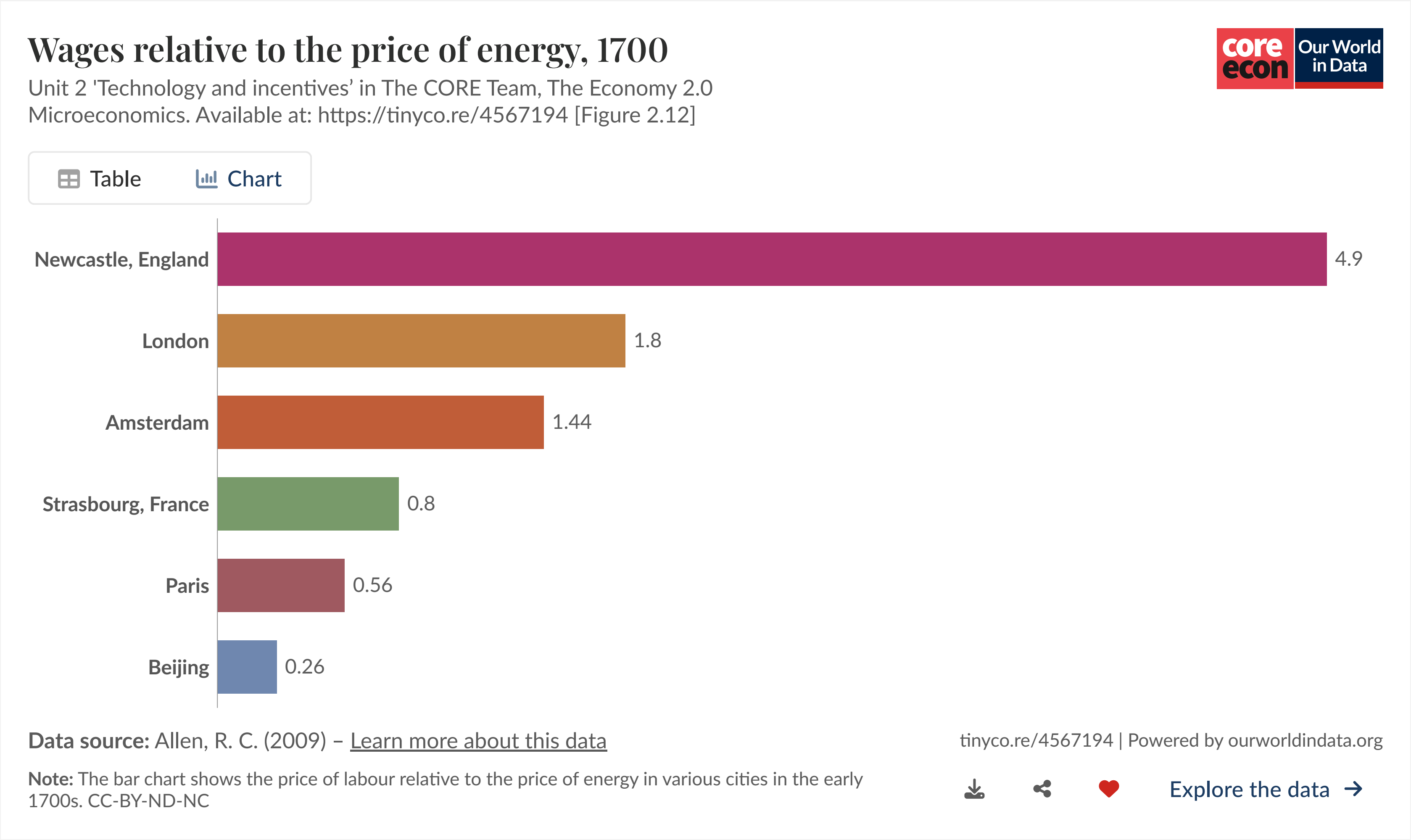


Cheap coal, expensive labour: The Industrial Revolution in Britain and incentives for new technologies

Spinning jenny: $\rightarrow$ A machine operated by 1 adult replaced 8 spinsters. By late XIXth century, 1,000 spinsters.

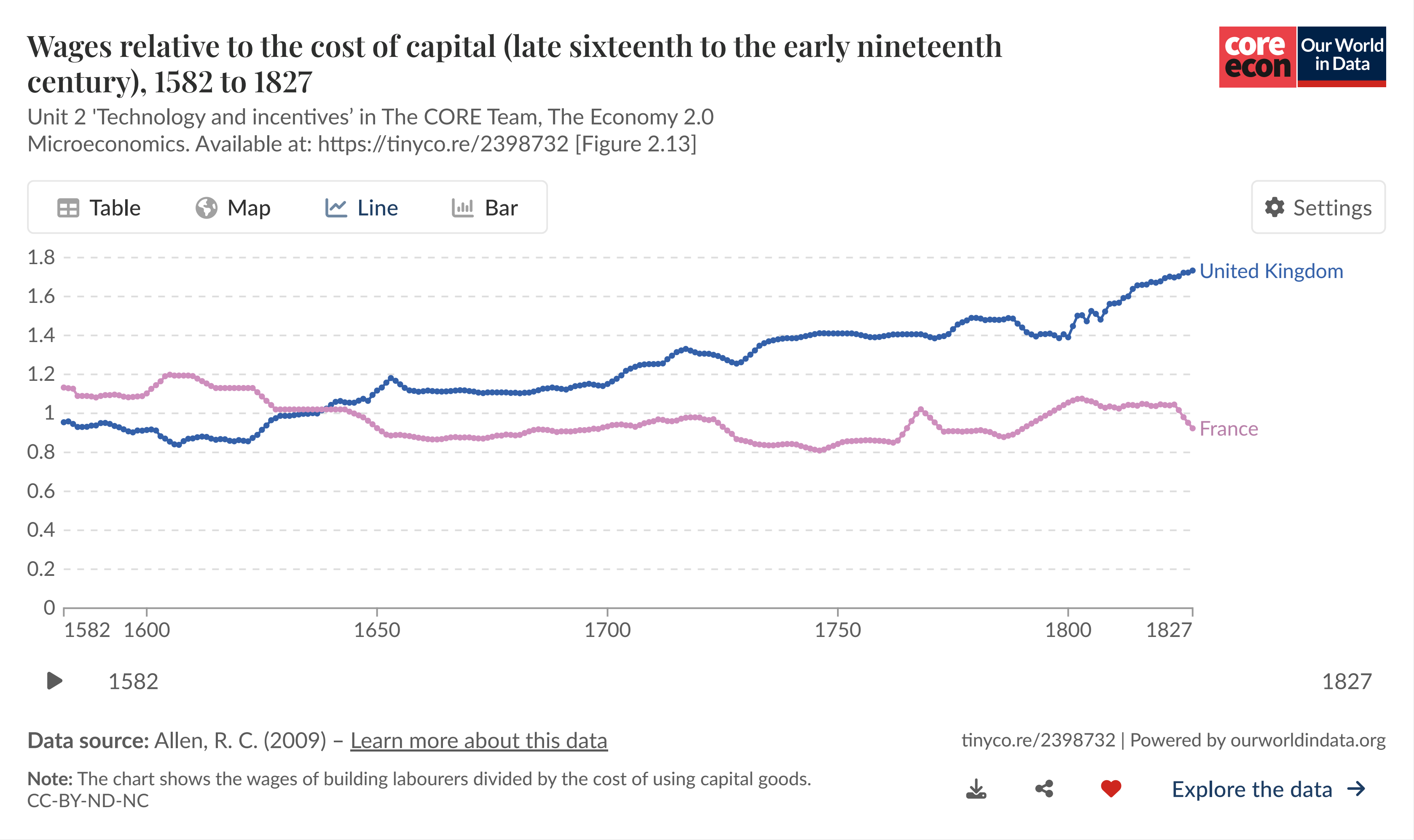
The Industrial Revolution in Britain and incentives for new technologies

The model suggests why people invent and adopt new technologies. But what does the evidence show?



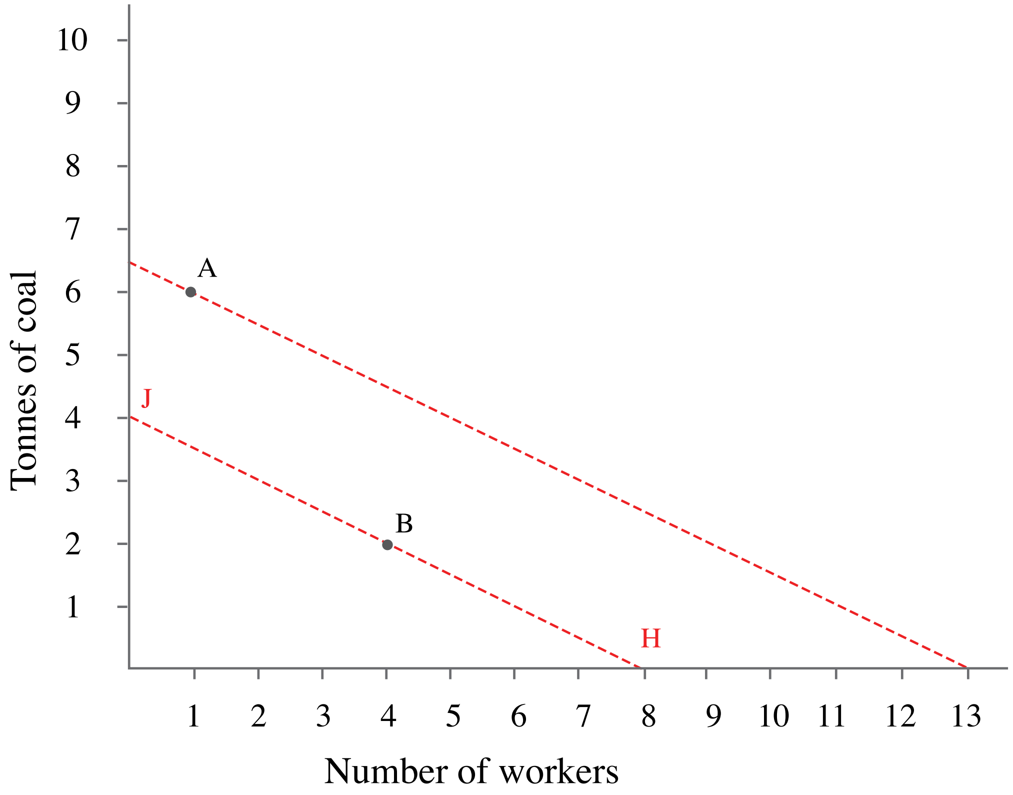
The Industrial Revolution in Britain and incentives for new technologies

The incentive to substitute machines for workers was rising in England, but not in France during this period.



The Industrial Revolution in Britain and incentives for new technologies

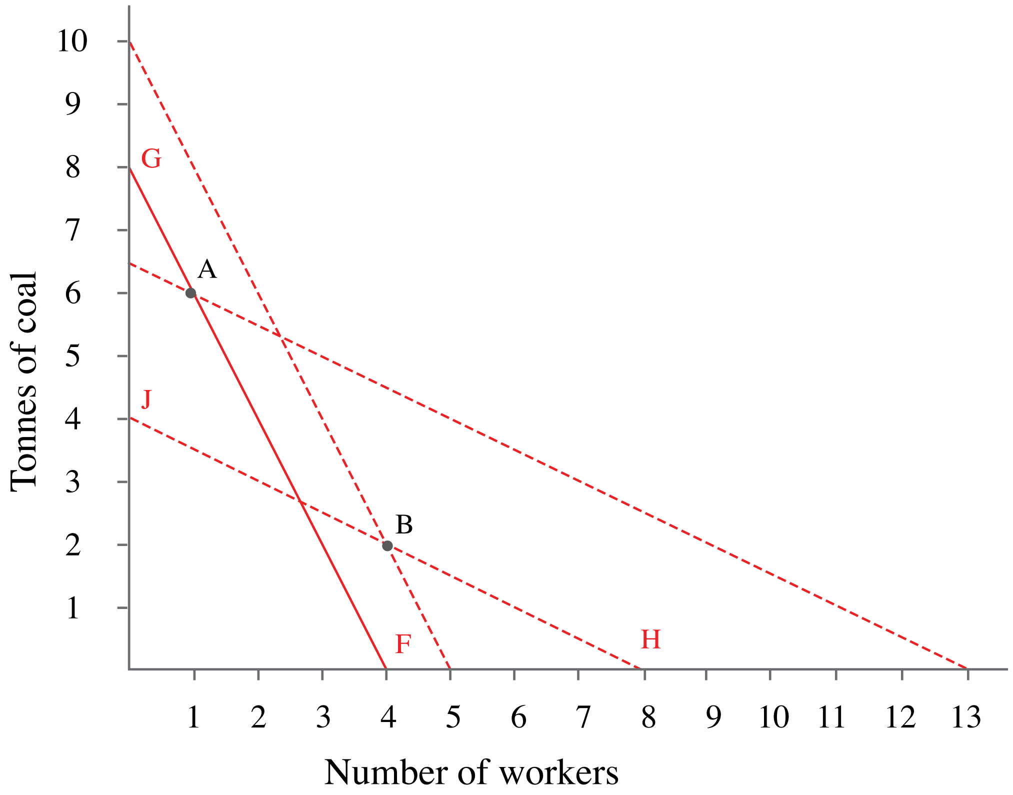
Technology choose depends on relative input prices



In the 1600s, the relative prices are shown by isocost line HJ. The B-technology was used. At those relative prices, there was no incentive to develop a technology like A, which is outside the isocost line HJ.

The Industrial Revolution in Britain and incentives for new technologies

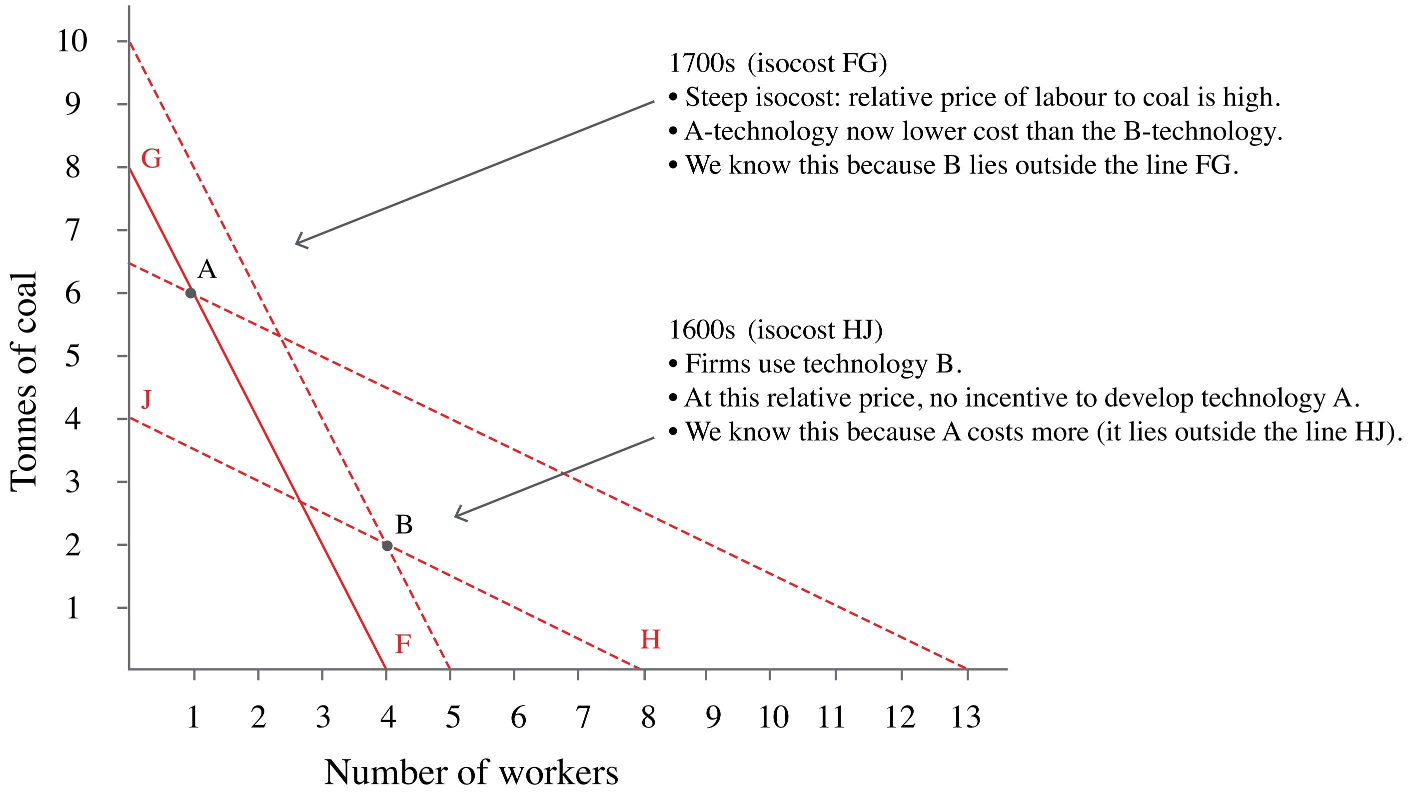
The combination of capacity to innovate and changing relative prices of inputs led to a switch to energy-intensive technology.



In the 1700s, the isocost lines such as FG were much steeper, because the relative price of labour to coal was higher. The relative cost was sufficiently high to make the A-technology lower cost than the B-technology.

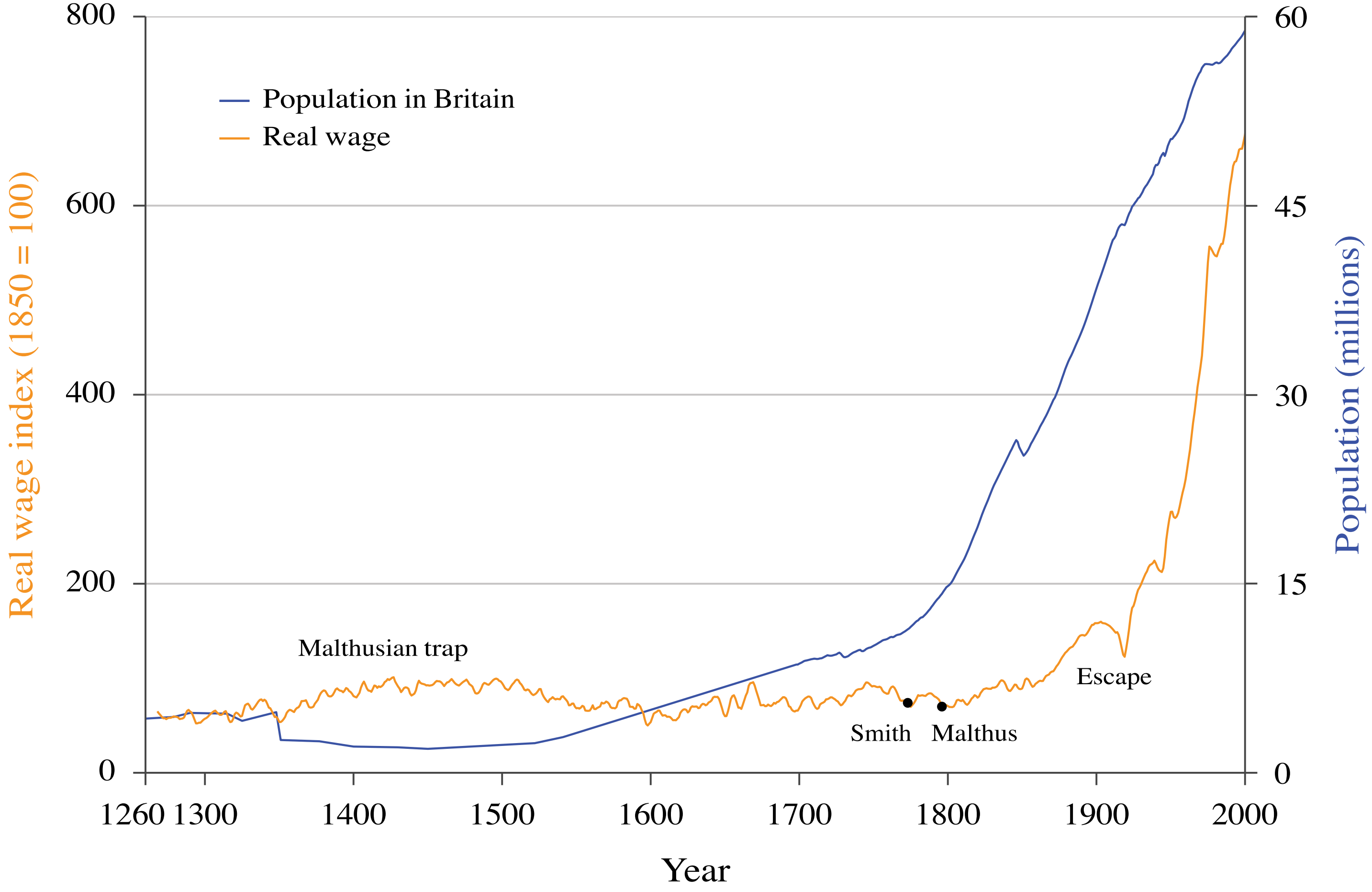
The Industrial Revolution in Britain and incentives for new technologies

The combination of capacity to innovate and changing relative prices of inputs led to a switch to energy-intensive technology.

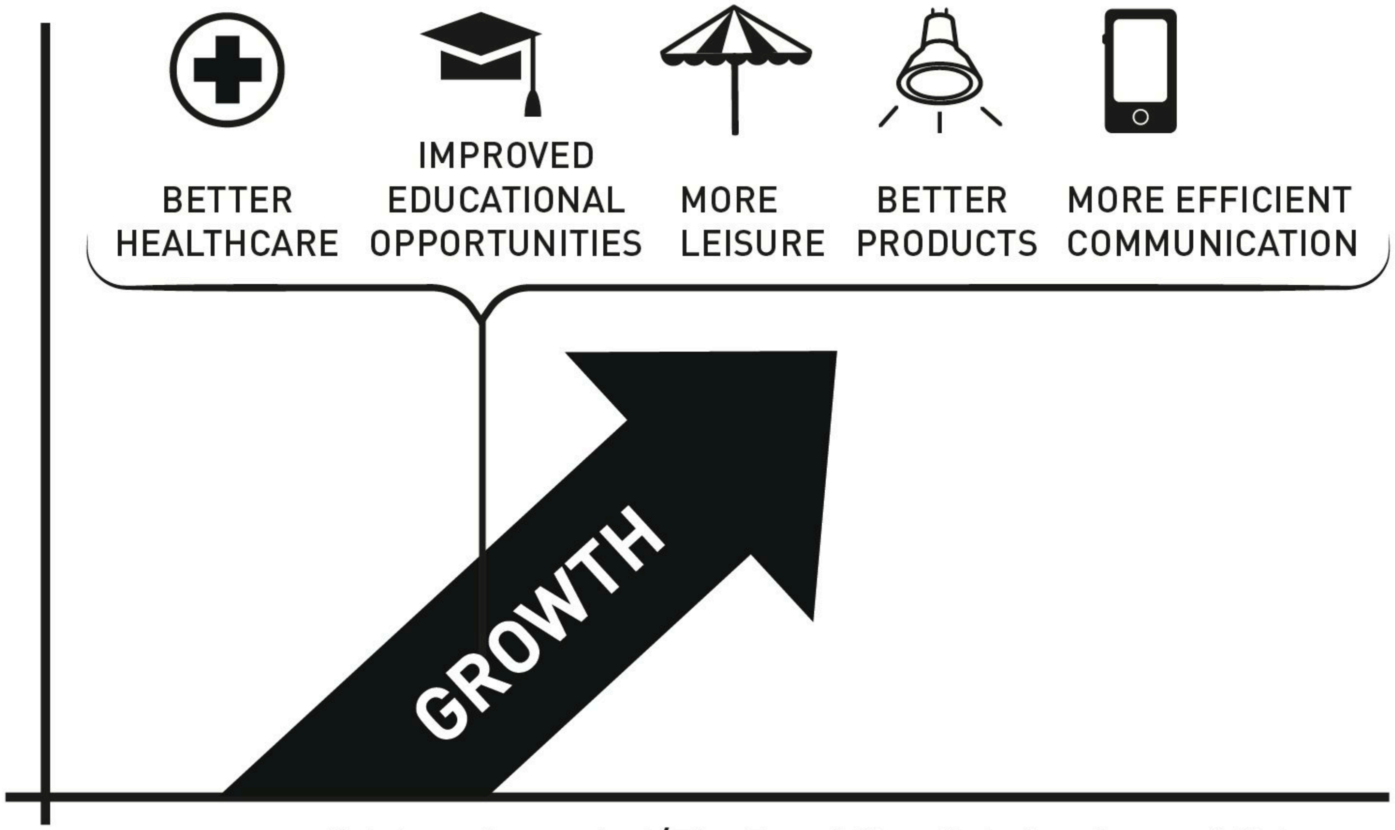


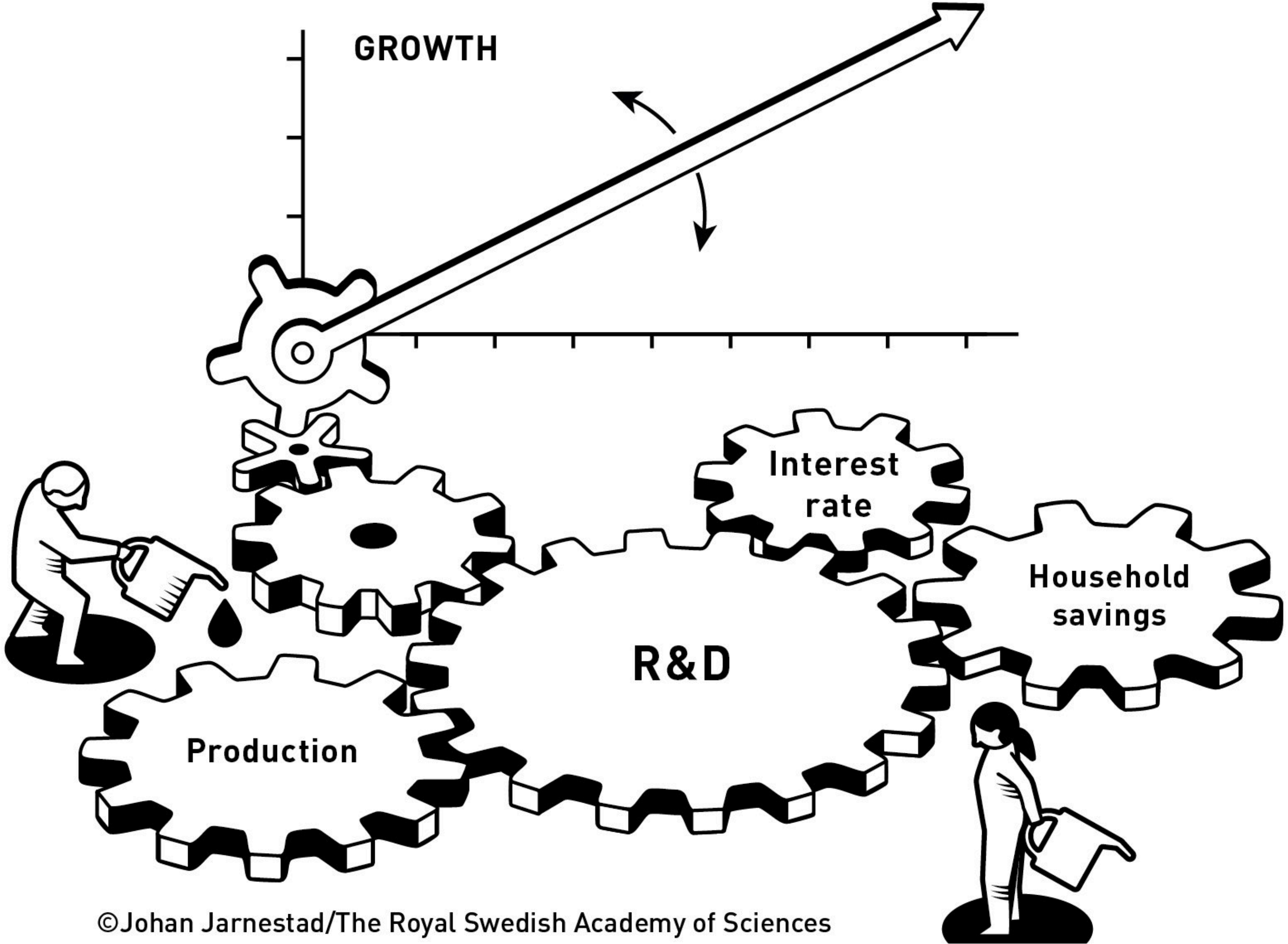
We know that when the relative price of labour is high, technology A is lower cost because the B-technology lies outside the isocost line FG.

The Industrial Revolution in Britain and incentives for new technologies

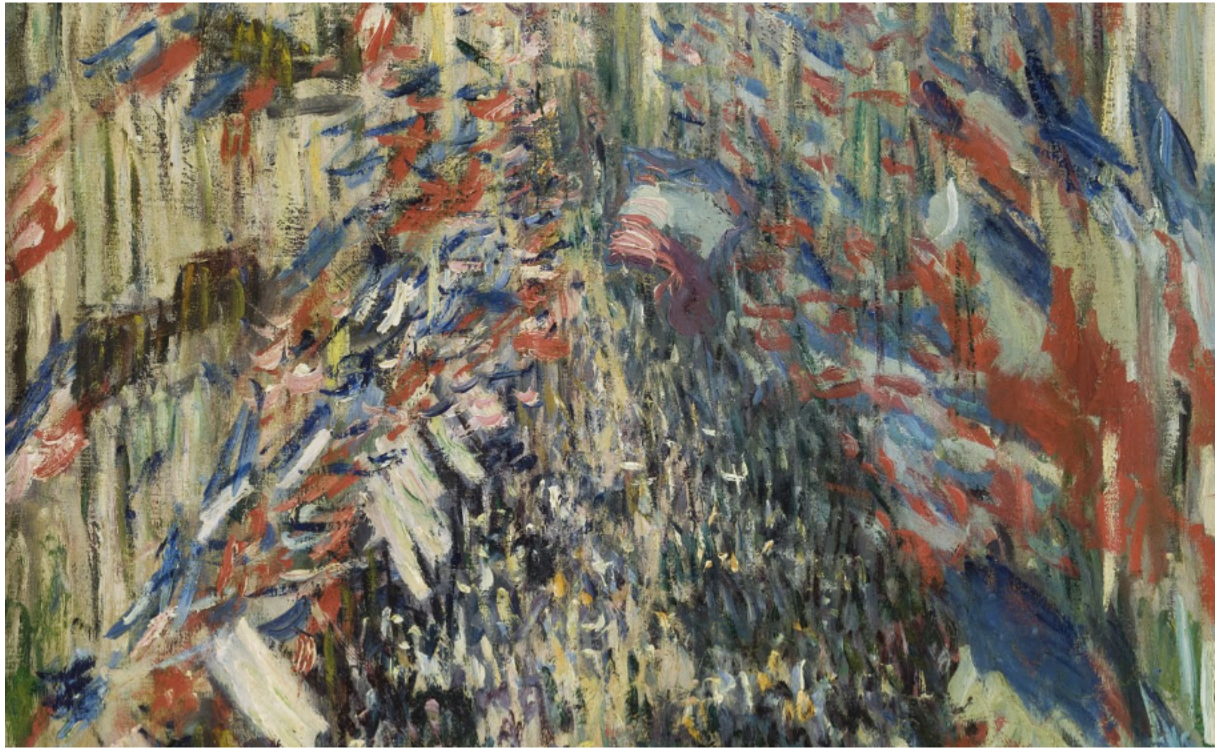


This is part of the explanation of the upward kink in the hockey stick. Explaining the long flat part of the stick is another story, requiring a different model.


BETTER HEALTHCARE
IMPROVED EDUCATIONAL OPPORTUNITIES
MORE LEISURE
BETTER PRODUCTS
MORE EFFICIENT COMMUNICATION
GROWTH


GROWTH
Interest rate
Household savings
R&D
Production

Economic models: How to see more by looking at less



La Rue Montorgueil, à Paris. Fête du 30 juin 1878. Claude Monet.

Economic models: How to see more by looking at less

Economic outcomes depend on millions of interactions between economic actors.

It would be impossible to understand the economy by describing every detail of how they act and interact.

We need to be able to stand back and look at the big picture. To do this, we use models.

Models *ingredients*:

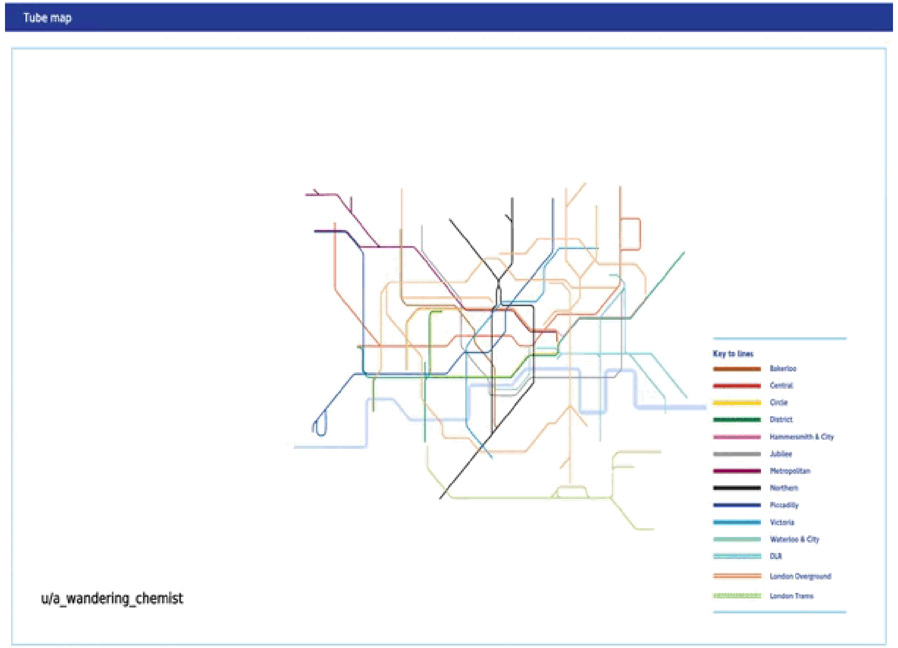
- **Incentives**: economic rewards or punishments, which influence the benefits and costs of alternative courses of action.
- **Equilibrium**: situation that is self-perpetuating. Something of interest does not change unless an external force is introduced that alters the model's description of the situation.
- **endogenous variables**: ‘generated by the model’. In an economic model, a variable is endogenous if its value is determined by the workings of the model (rather than being set by the modeller)
- **exogenous variables**: ‘generated outside the model’. In an economic model, a variable is exogenous if its value is set by the modeller, rather than being determined by the workings of the model itself.
- **Ceteris paribus**: simplification that involves "holding other things (in/outside the model) constant”.

Economic models: See more by looking at less

We need to be able to stand back and look at the big picture. They often use mathematical equations and graphs as well as words and pictures.

A good model:

- **Is clear**: it helps us better understand something important
- **Predicts accurately**: its predictions are consistent with evidence
- **Improves communication**: it helps us to understand what we agree (and disagree) about
- **It is useful**: We can use it to find ways to improve how the economy works

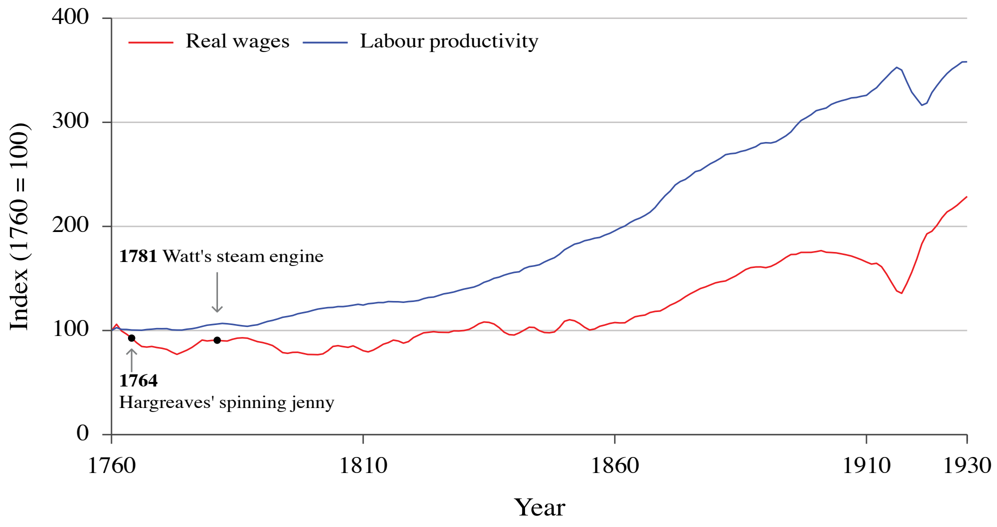


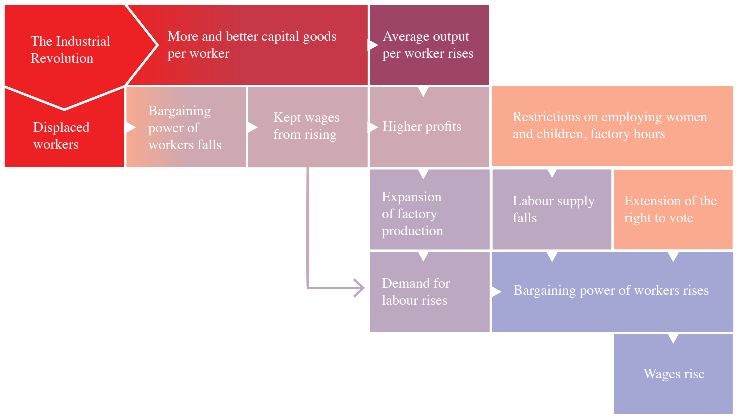
Bad models can result in disastrous policies. To have confidence in a model, we need to see whether it is consistent with evidence.

Markets, cheap calories, and cotton: The colonies, slavery, and the Industrial Revolution in Britain

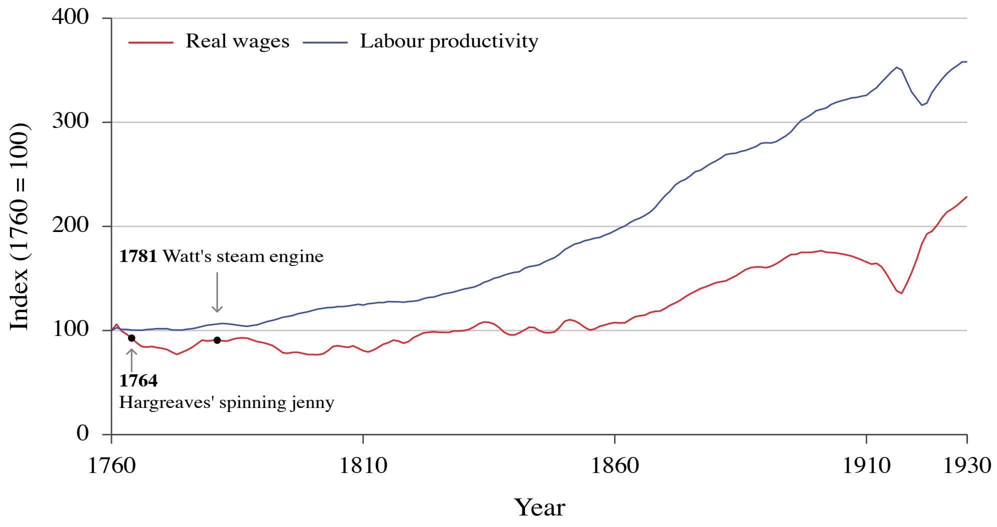
Growth: Escaping the Malthusian trap

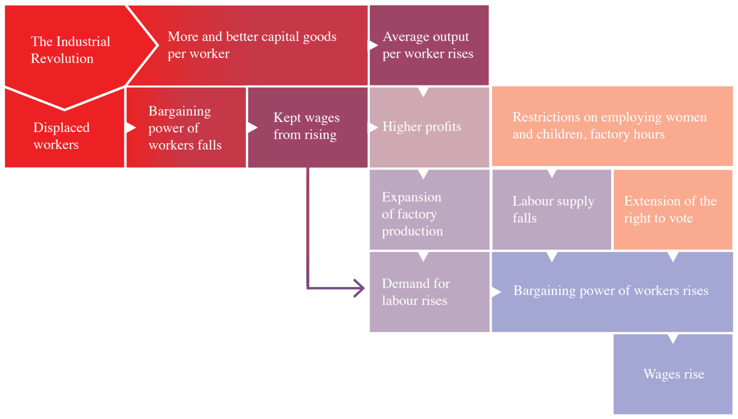
Growth: Escaping the Malthusian trap



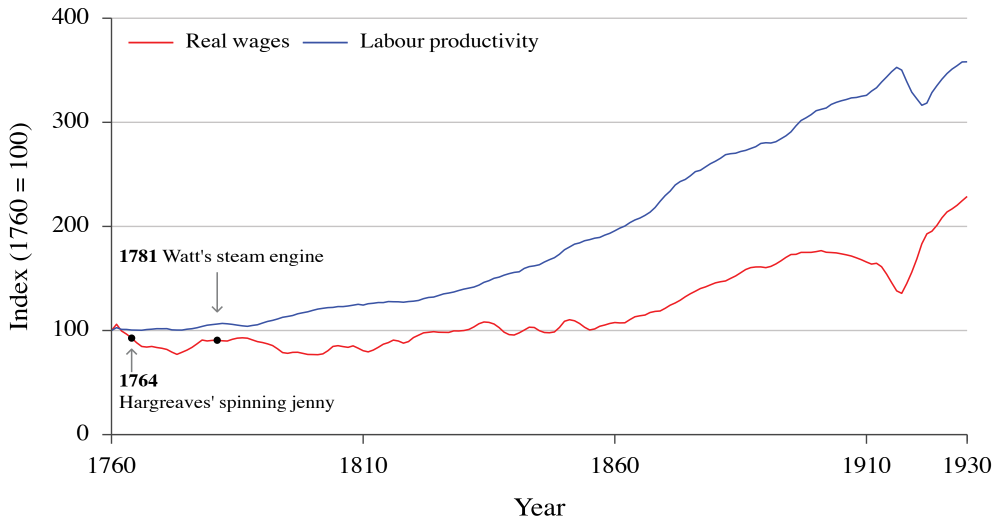


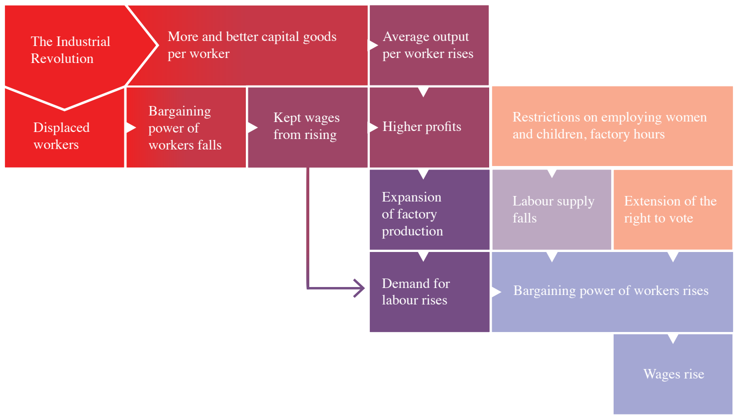
The story begins with technological improvements, such as the spinning jenny and the steam engine, that increased output per worker. Innovation continued as the technological revolution became permanent, displacing thousands of spinsters, weavers and farmers.



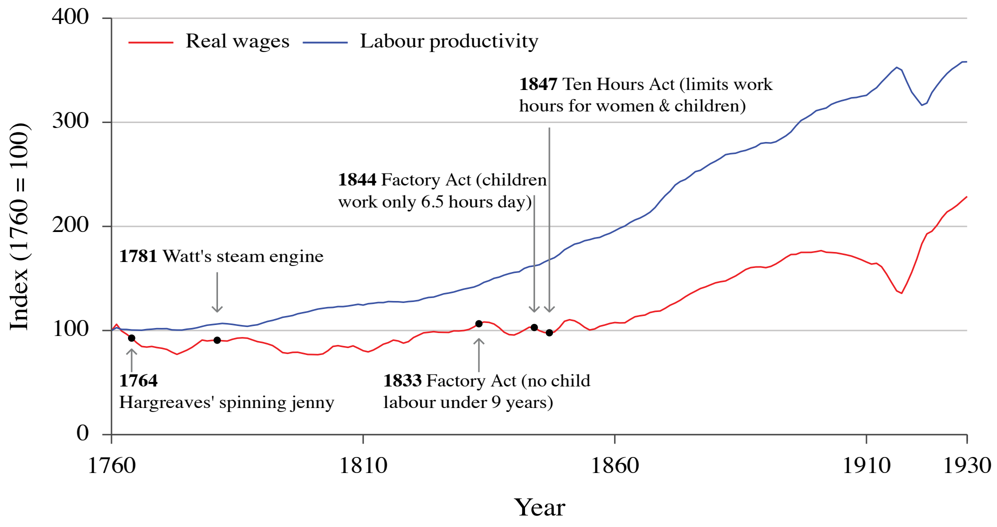


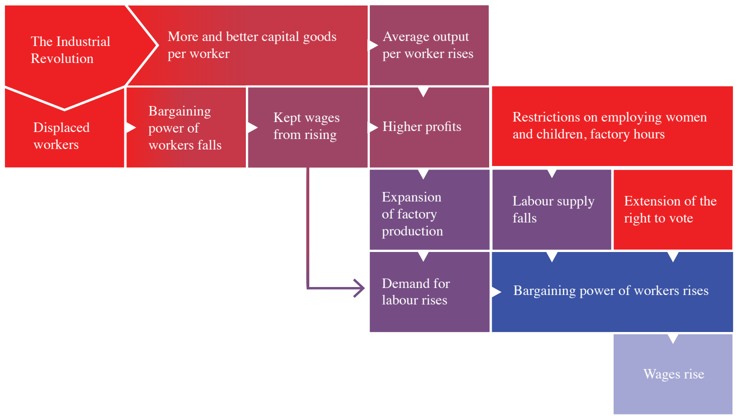
The loss of employment reduced workers’ bargaining power, keeping wages low, seen in the flat line between 1750 and 1830. The size of the pie was increasing, but the workers’ slice was not.



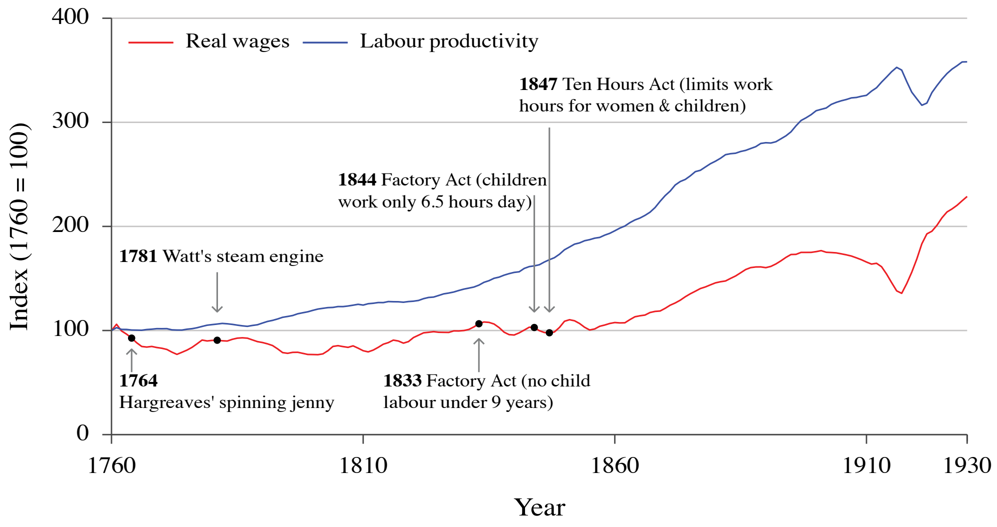


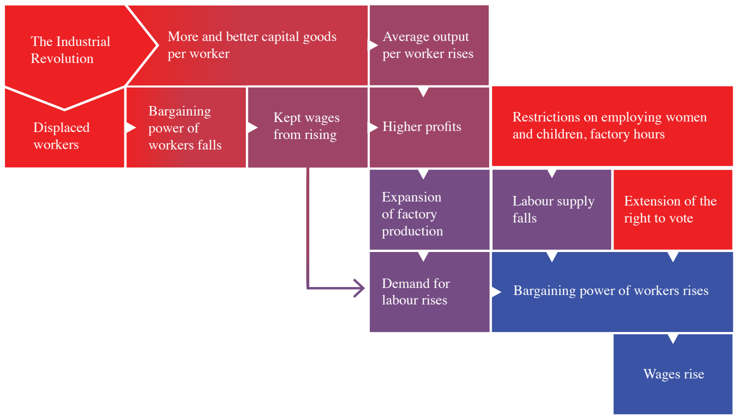
In the 1830s, higher productivity and low wages led to a surge in profits. Profits, competition, and technology drove businesses to expand. The demand for labour went up. People left farming for jobs in the new factories.





The supply of labour fell when business owners were stopped from employing children. The combination of higher labour demand and lower supply enhanced workers’ bargaining power.



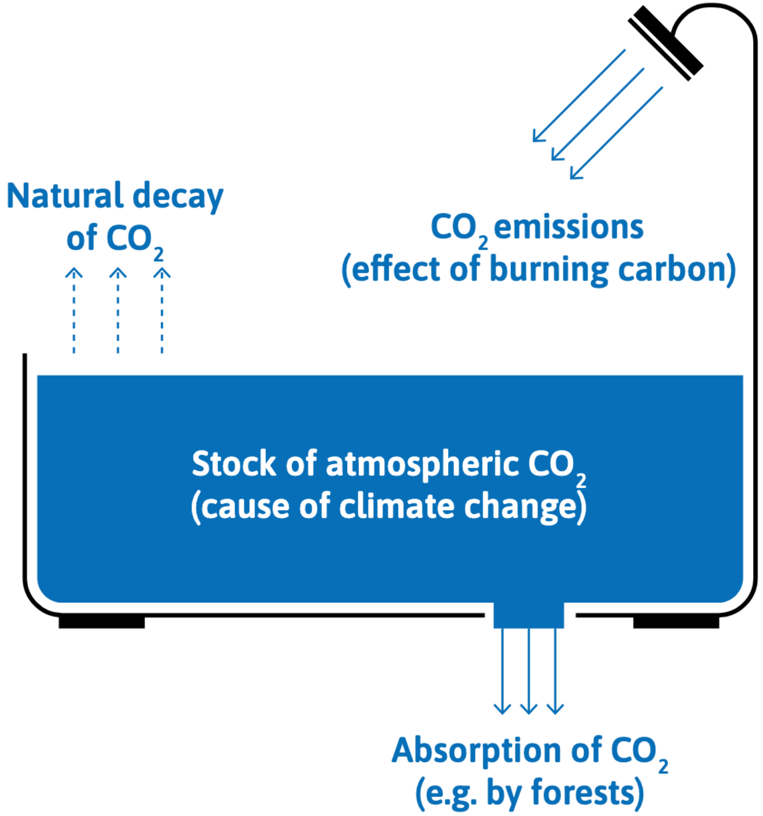


The power of working people increased as they gained the right to vote and formed trade unions. These workers were able to claim a constant or rising share of the increases in productivity generated by the permanent technological revolution.

Capitalism + carbon = hockey stick growth + climate change

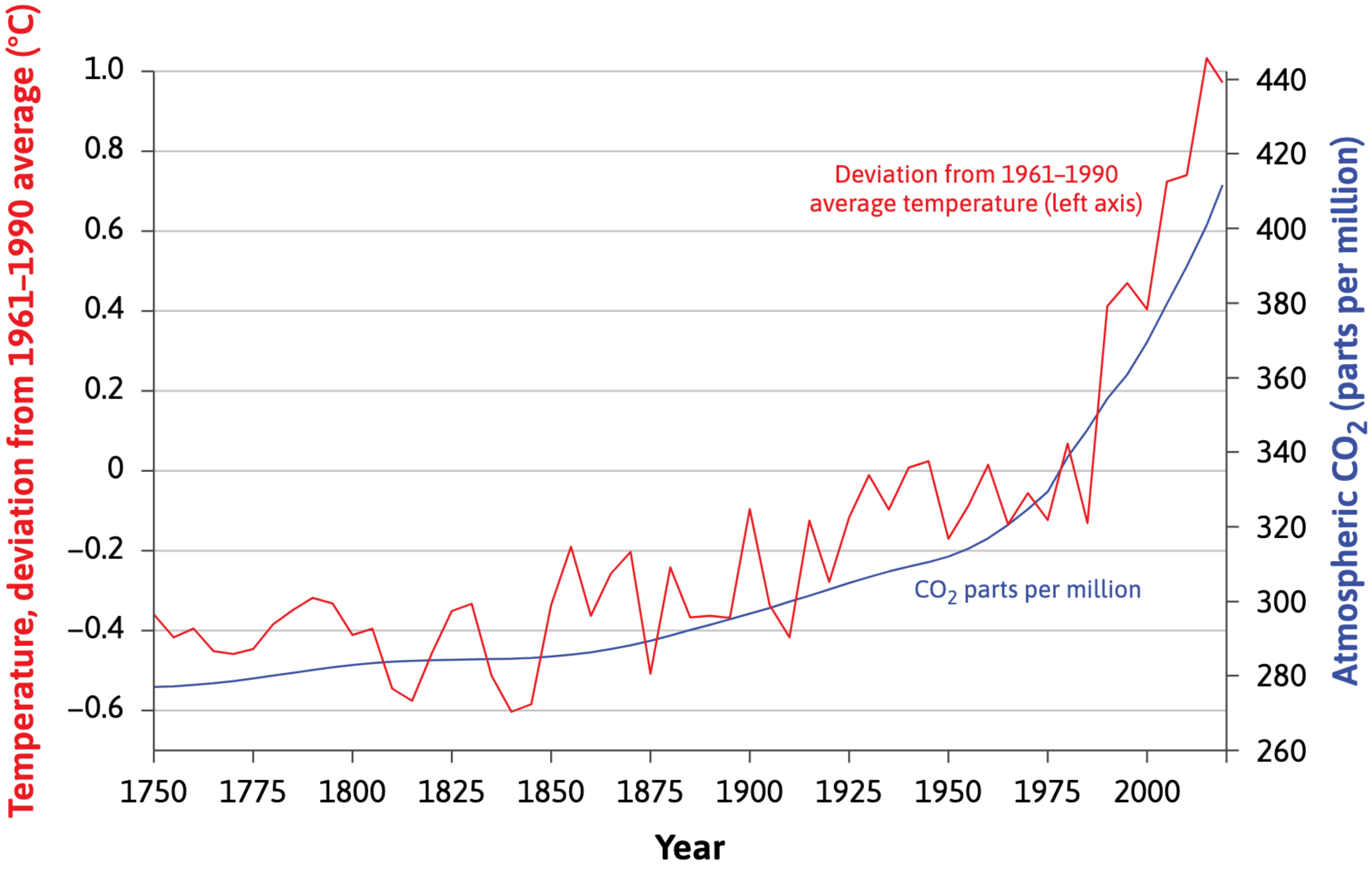
Capitalism + carbon = hockey stick growth + climate change

A bathtub model: the stock of atmospheric CO_2



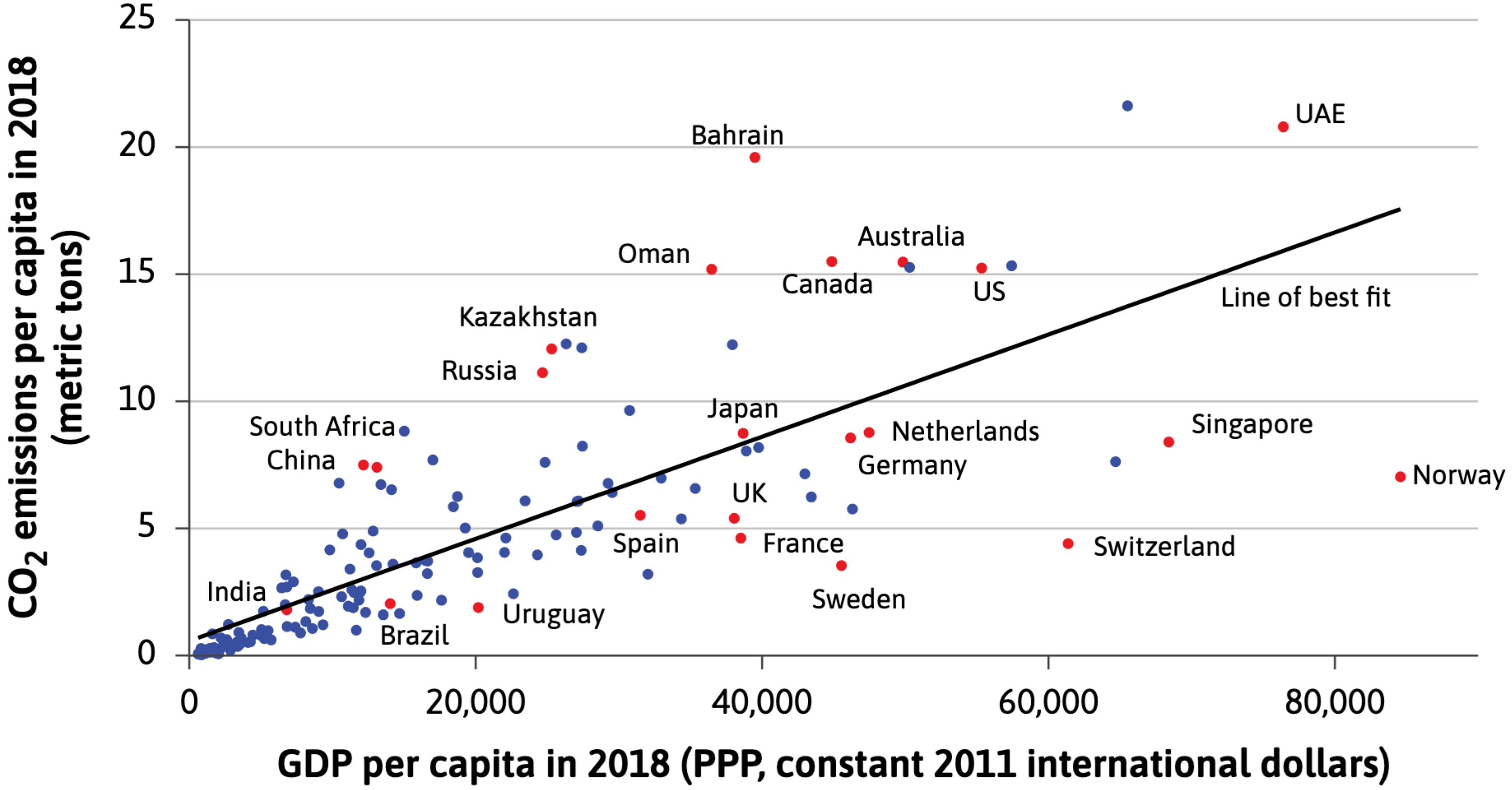
- **Stock**: amount of CO2 in the atmosphere
- **Flow**: amount being added per year

Capitalism + carbon = hockey stick growth + climate change



Global atmospheric concentration of carbon dioxide and global temperatures

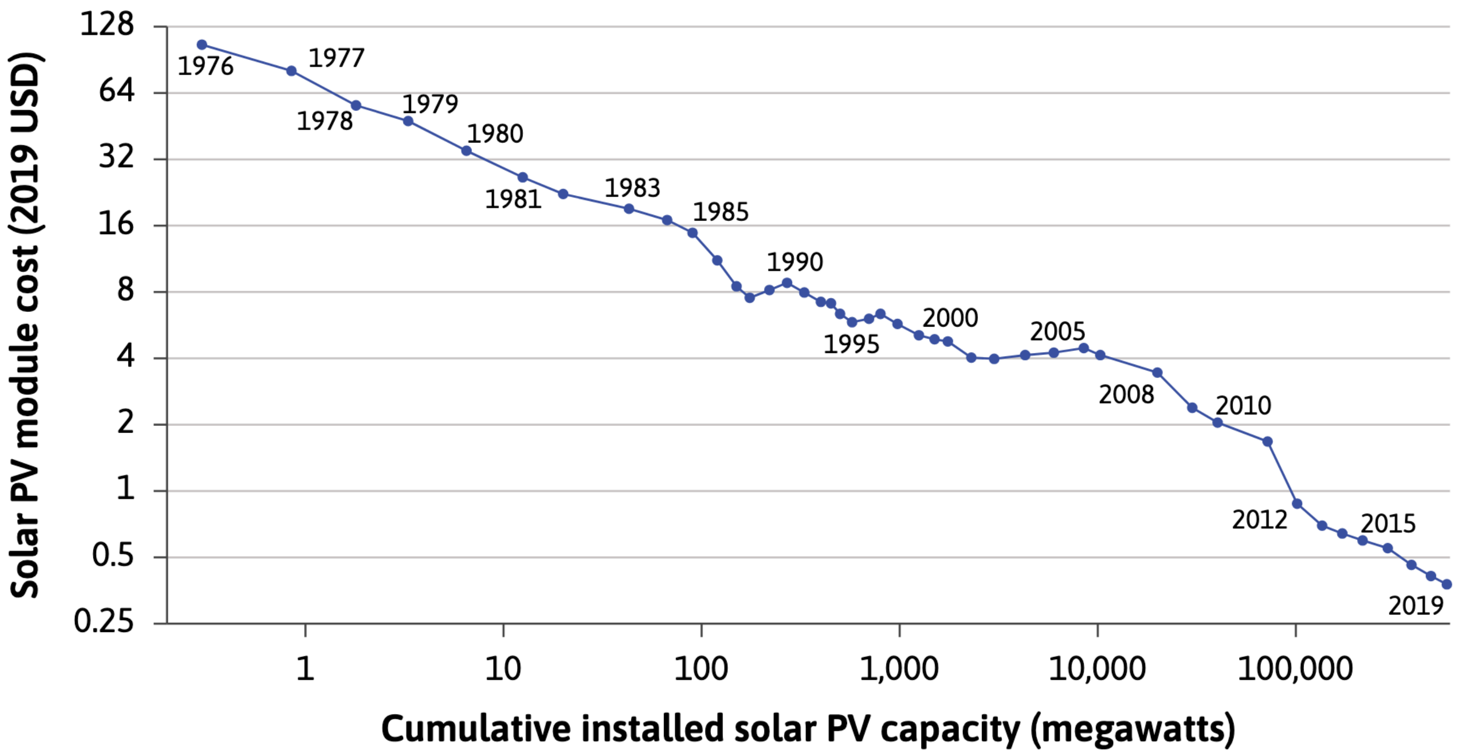
Capitalism + carbon = hockey stick growth + climate change



Carbon dioxide emissions are higher in richer countries

Capitalism + carbon = hockey stick growth + climate change

Can the technological innovation model help predict or explain aspects of climate change?



Capitalism + carbon = hockey stick growth + climate change

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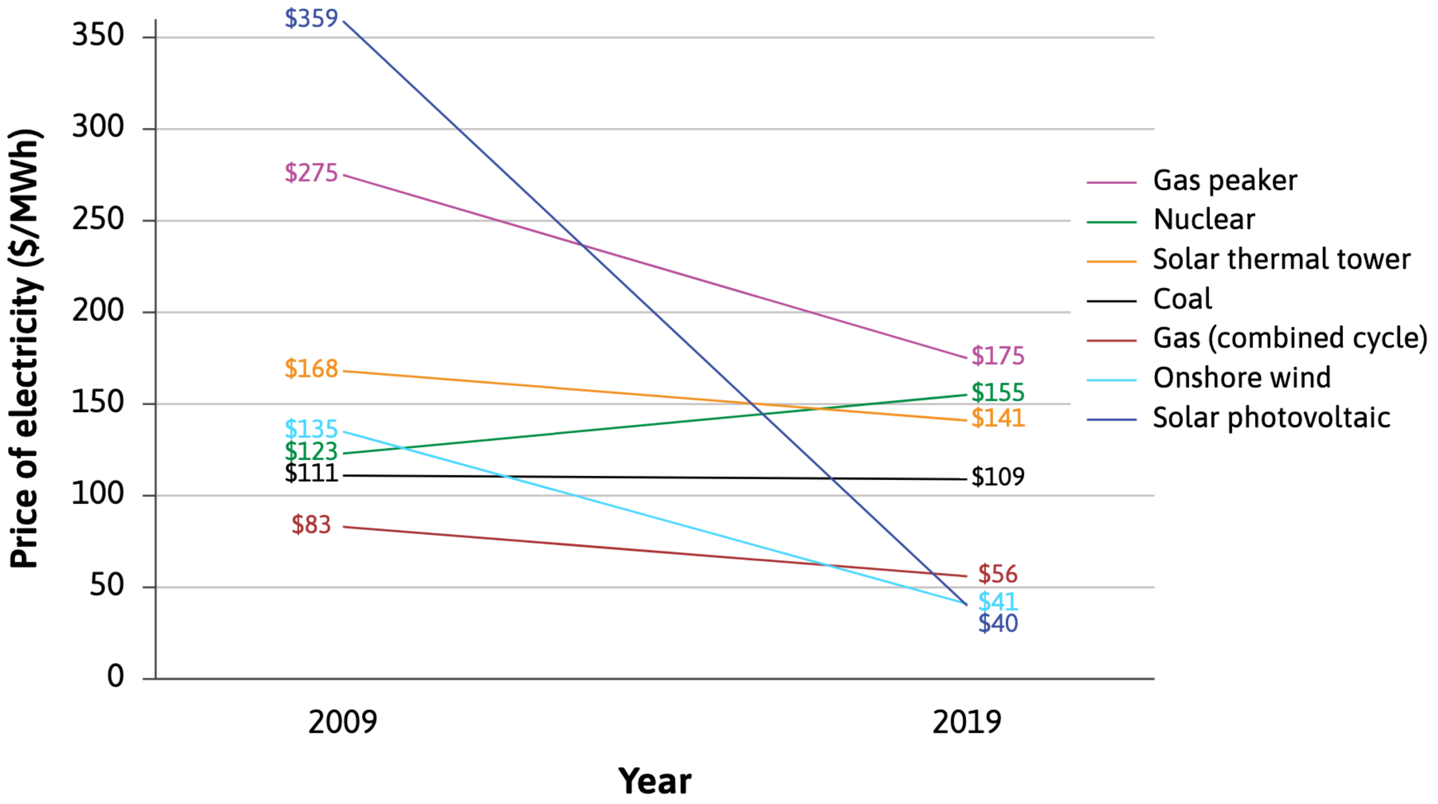
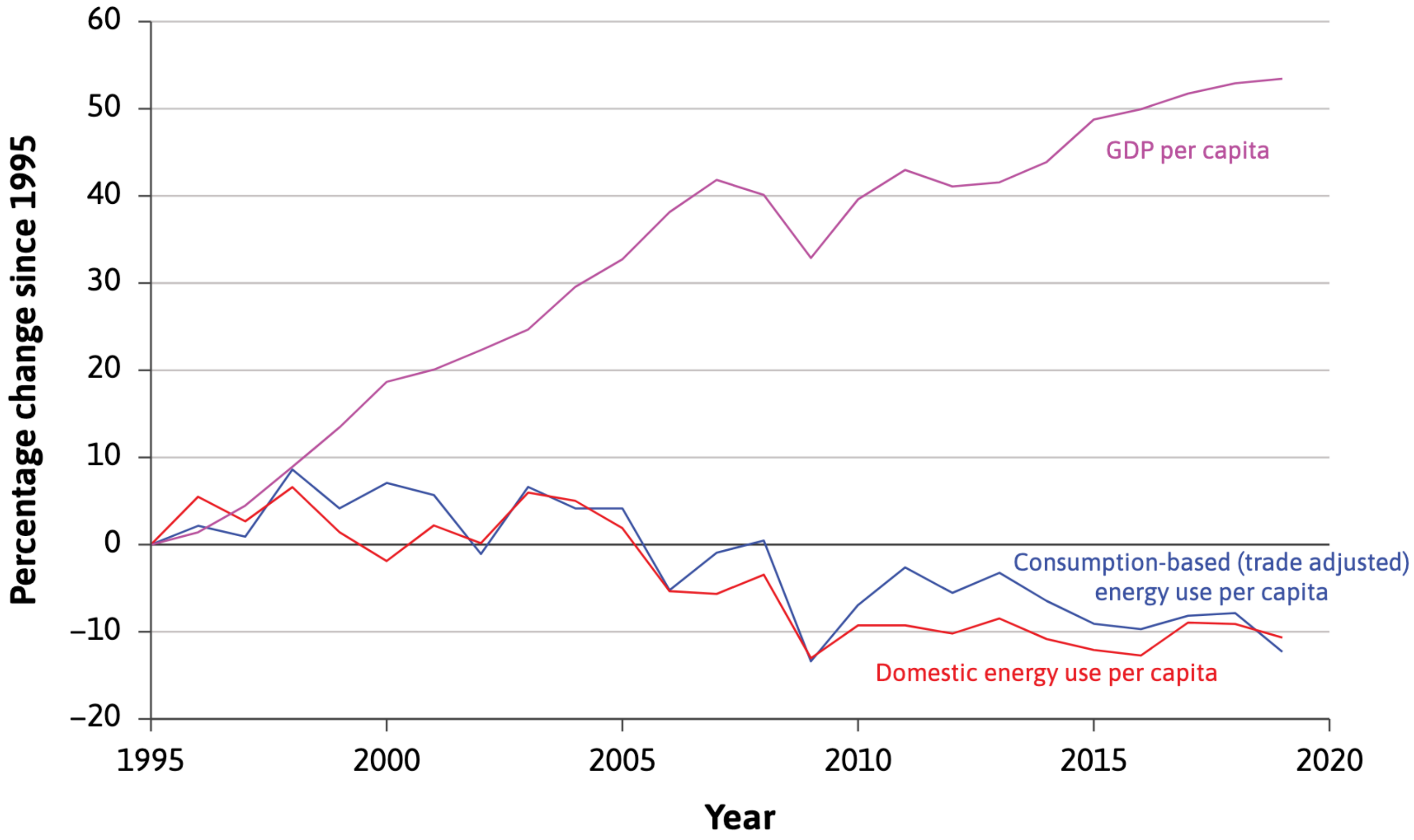


Capitalism + carbon = hockey stick growth + climate change

Can the technological innovation model help predict or explain aspects of climate change?



Seems like it does.

How good is the model? Economists, historians, and the Industrial Revolution

How good is the model? Economists, historians, and the Industrial Revolution

Why did the Industrial Revolution happen first in the 18th Century, on an island off the coast of Europe?

Many alternative explanations provided by historians, economic historians, and economists

- **Robert Allen**: relatively high cost of labour & cheap local sources of energy
- **Joel Mokyr**: Europe's scientific revolution and Enlightenment
- **David Landes**: political and cultural characteristics of nations as a whole
- **Gregory Clark**: cultural attributes such as hard work and savings
- **Kenneth Pomeranz**: abundance of coal and access to colonies

Economists can learn from historians, but historical arguments are often too imprecise for economic models. Conversely, historians may find economic models too simplistic.

(If you're interested in **drama, academic gossip, or debates** between historians and economists, I highly recommend this post.)


But it was so long ago. Why should I care?


THE SVERIGES RIKSBANK PRIZE
IN ECONOMIC SCIENCES IN MEMORY
OF ALFRED NOBEL 2025
Illustrations: Niklas Elmehed
Joel
Mokyr
"for having identified the
prerequisites for sustained
growth through
technological progress"
Philippe
Aghion
Peter
Howitt
"for the theory of sustained growth
through creative destruction"
THE ROYAL SWEDISH ACADEMY OF SCIENCES

Summary

Summary

- Rising productivity in 18th-century Britain broke the Malthusian trap, leading to higher wages and improved living standards.

- Firms adopted new technologies to reduce costs and increase profits, driven by incentives from relative prices and opportunity costs.

- Specialization and trade based on comparative advantage boosted productivity, supported by expanding markets and firm-based production.

-Cheap coal and high wages encouraged innovation in labour-saving, energy-intensive technologies, fueling industrial growth.

- Global connections—including access to colonial resources and slavery—underpinned Britain’s industrial expansion, while dependence on fossil fuels set the stage for modern climate challenges.